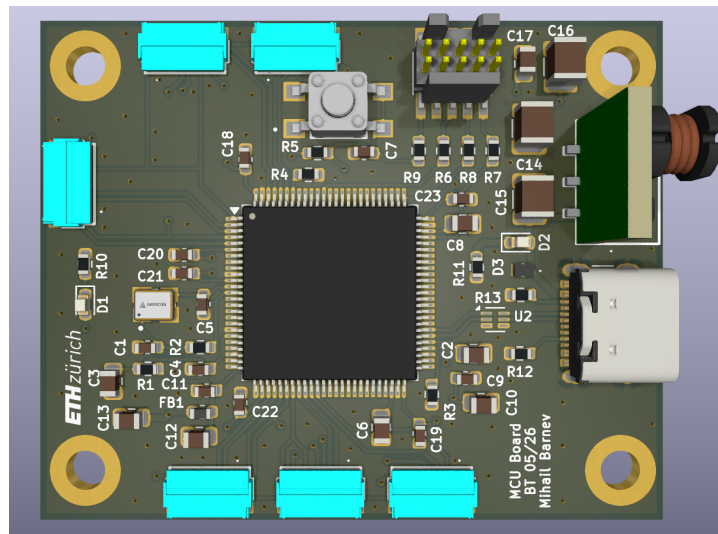


DEPARTMENT OF INFORMATION TECHNOLOGY AND
ELECTRICAL ENGINEERING

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Integration of Distributed dToF Sensors on a Quadrupedal Robot

Bachelor Thesis

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Abstract

Autonomous quadrupedal locomotion across uneven terrain often relies on exteroceptive depth data to construct local elevation maps. However, legged platforms operate under strict space, weight, and power constraints that limit their ability to maintain comprehensive visibility around the chassis. Lightweight and low-power Time-of-Flight (ToF) sensors offer a promising sensing approach to overcome these physical constraints. To improve spatial coverage for terrain perception, this thesis presents the hardware design and firmware development of a distributed depth-sensing system. The architecture integrates an array of six sensor nodes, each utilizing a VL53L8CX direct Time-of-Flight (dToF) multizone sensor to deliver localized grids of ranging data.

The primary objective of this work was to design and fabricate a custom MCU PCB functioning as the control hub to power and drive six sensor boards at peak operational capacity. Additionally, the sensor boards were reviewed and modified with the hardware components required to adapt them into the complete module. To manage the high-throughput requirements generated by the multi-sensor array (defined as the collective network of six independent sensor boards), a high-performance STM32F723 MCU was integrated, utilizing its on-chip USB High-Speed interface for fast data streaming to the robot's primary computer. The MCU board coordinates the sensor boards across two shared SPI buses (three sensors per bus), with each sensor configured to operate at maximum power capacity using a continuous 15 Hz frame rate and an 8×8 zone resolution.

Comprehensive electrical profiling verified that the board handles the full system load with high stability, demonstrating a clean linear power scaling profile from a 101.3 mA baseline up to a 400.5 mA maximum load under full operational stress. Furthermore, experimental testing validated the array's performance across diverse material textures and its resilience against mutual interference. The hardware and firmware successfully executed coordinated multi-node data acquisition as intended, demonstrating that a multiplexed dToF matrix array can provide a power-efficient exteroceptive sensing solution viable for future integration into resource-constrained robotic platforms.

Declaration of Originality

I hereby confirm that I am the sole author of the written work here enclosed and that I have compiled it in my own words. Parts excepted are corrections of form and content by the supervisor. For a detailed version of the declaration of originality, please refer to Appendix B

Mihail Barnev,
Zurich, June 2026

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List of Acronyms

CPU		Central Processing Unit
dToF		direct Time-of-Flight
EMI		Electromagnetic Interference
ESD		Electrostatic Discharge
FFC		Flexible Flat Cable
FoV		Field of View
FPC		Flexible Printed Circuit
FS		Full-Speed
GPIO		General-Purpose Input/Output
GPU		Graphics Processing Unit
HS		High-Speed
I2C		Inter-Integrated Circuit
LQFP		Low-Profile Quad Flat Package
MCU		Microcontroller Unit
PBL		Center for Project-Based Learning
PCB		Printed Circuit Board

List of Acronyms

PHY		physical-layer
PPK2		Power Profiler Kit II
SLAM		Simultaneous Localisation and Mapping
SPAD		Single Photon Avalanche Diode
SPI		Serial Peripheral Interface
SRAM		Static Random-Access Memory
ToF		Time-of-Flight
ULPI		UTMI+ Low Pin Interface
USB		Universal Serial Bus
VCSEL		Vertical-Cavity Surface-Emitting Laser

Introduction

Autonomous quadrupedal robots are highly versatile platforms engineered to navigate complex environments [1]. To perform stable locomotion and plan safe paths, these platforms must construct an accurate map of their immediate surroundings [2, 3]. This map is typically built using 2.5D elevation mapping, which tracks terrain heights in a grid around the moving robot. Because path planners rely entirely on this map to navigate, the robot’s stability depends on the speed, placement, and accuracy of its onboard depth sensors [4].

Traditionally, dense exteroceptive depth mapping has relied on bulky hardware solutions, such as stereoscopic cameras, spinning LiDARs, or active 3D ToF range imagers. Although these conventional LiDAR and vision sub-systems provide fine-grained geometric point clouds, they introduce engineering bottlenecks on compact mobile machines. Processing high-resolution spatial frames requires power-hungry computational pipelines that consume vital processing cycles on the robot’s primary computer, thereby severely shortening battery runtime. Furthermore, these bulky sensors are typically concentrated in the front or rear chassis, leaving blind spots around the underbelly and along the lateral flanks of the quadruped framework.

To overcome these spatial limitations, recent research at the Center for Project-Based Learning (PBL) demonstrated that incorporating a supplementary ToF configuration can support robust legged locomotion in rough terrain [1]. Additionally, it was shown that by positioning miniature depth sensors to cover the blind spot beneath the chassis, this approach maximizes local spatial coverage without introducing physical bulk. This cuts down mapping errors so the robot can walk safely without relying on heavy visual-inertial odometry. However, building a distributed module using multiple multi-zone dToF sensors introduces distinct hardware and firmware integration challenges. Coordinated data acquisition from a multi-node array running concurrently requires an architecture capable of reliable multi-channel communication. This thesis builds upon this foundation by

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developing and validating the synchronized hardware and firmware required for such a multi-sensor network.

1.1. Objective

This project builds on the exteroceptive elevation mapping frameworks and reinforcement learning locomotion paradigms introduced by the Robotic Systems Lab (RSL) and the PBL [2, 1, 3, 4]. The core objective of this work is to design and experimentally validate a dToF perception module to eliminate the underbelly and lateral blind spots in a quadrupedal robot. The explicit engineering milestones include:

- Designing from the ground up and fabricating a high-performance Printed Circuit Board (PCB) centered around an STM32F723 Microcontroller Unit (MCU) to coordinate a distributed network of six sensor units (the MCU board).
- Reviewing the schematics of the sensor board (designed by Dr. Christian Vogt) and executing the hardware modifications required to guarantee connectivity of the full module (MCU board & sensor boards).
- Developing a firmware architecture to drive at their peak operational limits all six VL53L8CX sensors simultaneously over two concurrent Serial Peripheral Interface (SPI) buses.
- Integrating an on-chip Universal Serial Bus (USB) High-Speed (High-Speed (HS)) communication channel to stream the proximity point clouds.
- Establishing a profiling workflow to evaluate the module’s power consumption under full operational load and demonstrating its performance across diverse target materials and interference tests.

1.2. Outline

- **Chapter 2: Related Work** reviews the academic literature on robotic elevation mapping and different sensing architectures, and highlights prior empirical work on the characterization of multi-zone dToF sensors.
- **Chapter 3: Background** establishes the theoretical operating principles of direct Time-of-Flight sensing and details the behavior of the VL53L8CX sensor. Further, it outlines the architecture of the target quadrupedal robot and reviews the STMicroelectronics STM32 portfolio.

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- **Chapter 4: Hardware Design and Firmware Implementation** details the complete electronic development workflow, presenting the schematic design and PCB layout routing for the boards, the component selection trade-offs, and the configuration of the firmware.
- **Chapter 5: Results** presents the empirical benchmarks collected during laboratory trials, focusing on sensor testing and system power profiling metrics captured via a Nordic Power Profiler Kit II (PPK2).
- **Chapter 6: Discussion and Design Limitations** delivers an evaluation of hardware limitations, detailing assembly challenges, layout space optimization, and manual hardware modifications, and analyzes the results from Chapter 5.
- **Chapter 7: Conclusion and Future Work** summarizes the milestones achieved throughout this thesis and introduces future work for integrating the sensing module onto the robot chassis.

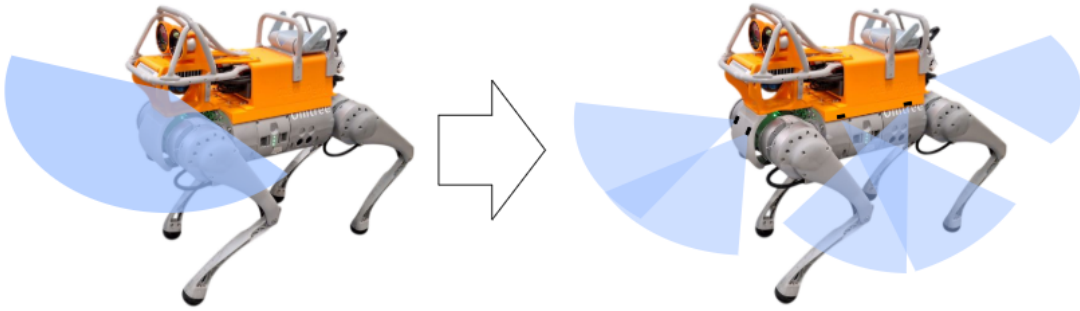


Figure 1.1.: Overview of the proximity sensing paradigm shift: from the localized coverage profile of a single primary sensor layout to a configuration utilizing a distributed network of multiple dToF sensors to provide proximity tracking around the whole robot

Chapter 2

Related Work

Evaluating the boundaries of a distributed exteroceptive sensor array requires a careful analysis of how contemporary literature balances mapping architecture, space constraints, and multi-zone distance accuracy.

Perception Legged robotic navigation frameworks traditionally process exteroceptive distance points inside a local terrain map that shifts dynamically alongside the machine’s physical coordinates [2]. These frameworks utilize a 2.5D elevation map where a 2D grid stores the height information of the ground for each cell. Compared to dense 3D voxel structures, this representation accurately captures rough terrain geometry while maintaining low memory requirements. This efficiency ensures that the local map updates rapidly, isolating geometric uncertainties from the low-level locomotion controller.

Spatial Coverage and Physical Constraints Modern elevation mapping frameworks provide excellent terrain reconstructions for legged systems. However, expanding the robot’s spatial view using high-resolution depth cameras or large LiDAR sensors introduces physical limitations [1]. These vision systems are often bulky and difficult to distribute effectively around a compact quadruped chassis. Furthermore, attempting to expand spatial coverage by mounting auxiliary visual configurations introduces additional weight and takes up limited space on chassis layouts. Therefore, the primary challenge is not a lack of onboard computing power, as modern legged platforms often incorporate high-performance GPUs, but rather the physical constraints of size, weight, and sensor placement.

Distributed Sensing Approaches To address the volume and placement constraints of large depth cameras, prior work in the PBL investigated mounting an additional rear-facing 3D indirect ToF camera, PMD Flexx2, to monitor vulnerable spaces beneath the chassis. This investigation demonstrated that an auxiliary depth camera successfully improves elevation mapping accuracy and preserves traversal performance in rough terrain [1, 3]. By extension, scaling this concept from a single camera to a network of

2. Related Work

multiple, distributed, miniature ToF sensors offers significant architectural benefits. Unlike cameras, these multi-zone sensors are small and lightweight. By positioning several of these miniature sensors around the perimeter of the quadruped, the system achieves wide spatial coverage and eliminates critical blind spots underneath the belly without adding unnecessary weight or physical bulk to the robot frame. The application of low-overhead proximity data has been mirrored across other resource-constrained autonomous platforms within the PBL framework, notably by Müller et al., where multi-zone ToF sensors were successfully deployed to achieve robust obstacle avoidance for autonomous miniaturized UAVs without requiring heavy on-board computation [5].

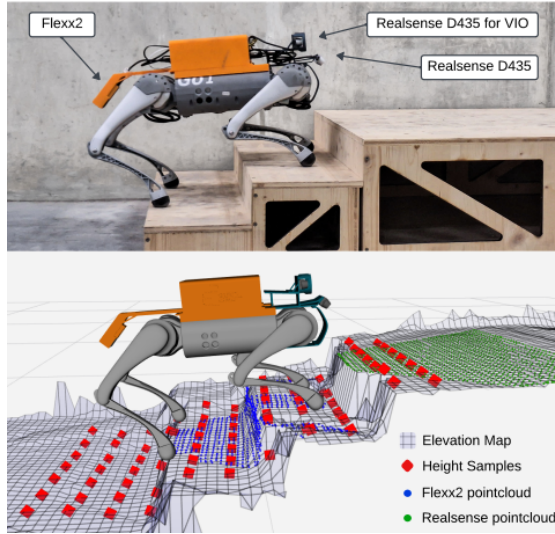


Figure 2.1.: Stair climbing deployment, mapping the terrain. The bottom visualization illustrates how sparse proximity point clouds are integrated into the robot height sampling policy. The figure is reused from the original paper [1].

Sensor Characterization As the proposed architecture coordinates six independent multi-zone sensors, their systematic characterization by Caroleo et al. at the University of Oxford provides essential data for this family of devices. Beyond distance profiling, their work demonstrated that dToF sensors could be used to compute complex state metrics, including full-body self-localization and shape estimation within constrained workspaces [6]. The findings detailed in [7] map the operational profiles that establish the performance boundaries for these VL53L5CX chips:

- **Thermal Stabilization:** Caroleo et al. discovered that continuous data streaming triggers a rapid increase in component temperature from room temperature to a steady-state value of $51.61 \pm 1.70^\circ\text{C}$. This thermal rise induces an exponential drift of the measurement from the ground truth during the first 15 minutes of device uptime before stabilizing. This established baseline highlights the necessity of implementing active temperature compensation mechanisms to ensure data integrity during continuous operation.

2. Related Work

- **Resistance to Ambient Illumination:** Their systematic testing demonstrated that standard indoor office illumination levels (from 0.5 lux up to 270 lux) exert a completely negligible effect on depth estimation accuracy and standard deviation. However, severe external infrared interference (such as a 3 klux incandescent source) floods the receiver array, causing data rejection rates to spike up to 45%. This sets a clear operational limit for the reliability of the array under severe ambient infrared noise.
- **Vulnerability to Dark and Absorptive Surfaces:** Ranging performance degrades significantly when encountering dark materials that absorb the 940 nm near-infrared light emission. When tracking a matte black foam target beyond a 400 mm distance, the reduced photon return causes standard deviation noise spikes of nearly 50 mm. Furthermore, on highly absorbent black vinyl surfaces, data corruption causes the valid measurement rate to drop to just 18.8% at a 25 cm distance and 34.4% at a 60 cm distance.
- **Receiver Architecture:** The underlying 8×8 grid integrates a Single Photon Avalanche Diode (SPAD) array to split returning signals into individual optical channels from a single emitter pulse. This enables parallel zone extraction, but highly reflective targets can induce subtle optical blur or interference artifacts, in which reflections bleed slightly into the distance calculations of adjacent tracking zones.
- **Angular Sensitivity and Surface Orientation:** Testing across varied target orientations revealed that tracking accuracy exhibits clean tolerances within small alignment angles, but measurement reliability declines as the target surface tilts away from the sensor face. This degradation is caused by the increased orientation angle, which reduces the effective surface area and thereby limits the number of photons reflected back to the SPAD array. This behavior justifies the deployment of a distributed array of sensors pointing in multiple directions to reliably cover angled obstacles.
- **Corner Deviations:** An inherent hardware trait of the matrix layout is that individual zone measurements deviate from a best-fitting plane within a tight 0.5 mm to 2.0 mm range. However, this noise is not perfectly uniform across the grid. The zones located at the corners of the 8×8 matrix suffer slightly higher standard deviations than the central zones because the alignment of the lens array collects fewer returning photons at the boundaries of the Field of View (FoV).

2. Related Work

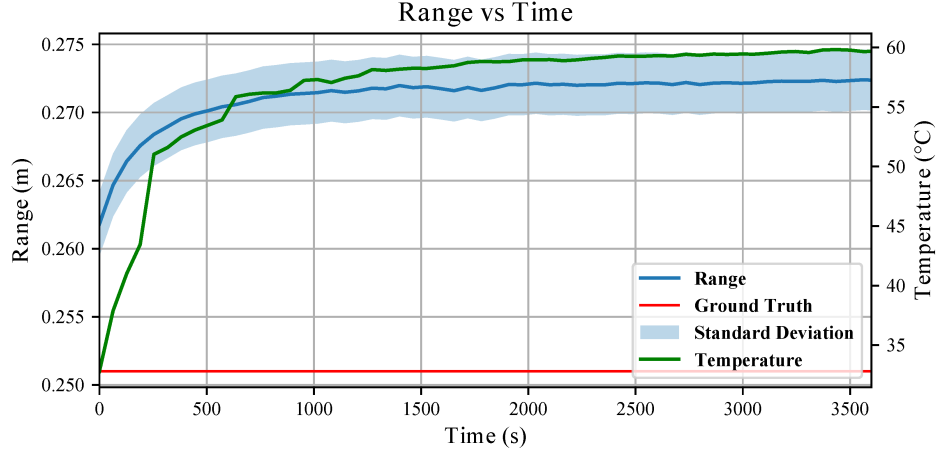


Figure 2.2.: Empirical sensor drift over 1-hour experiment. The plot shows the exponential rise and steady-state stabilization of the component temperature. Simultaneously, it maps the corresponding range measurement drift away from the static ground truth over time. The figure is reused from the original paper [7].

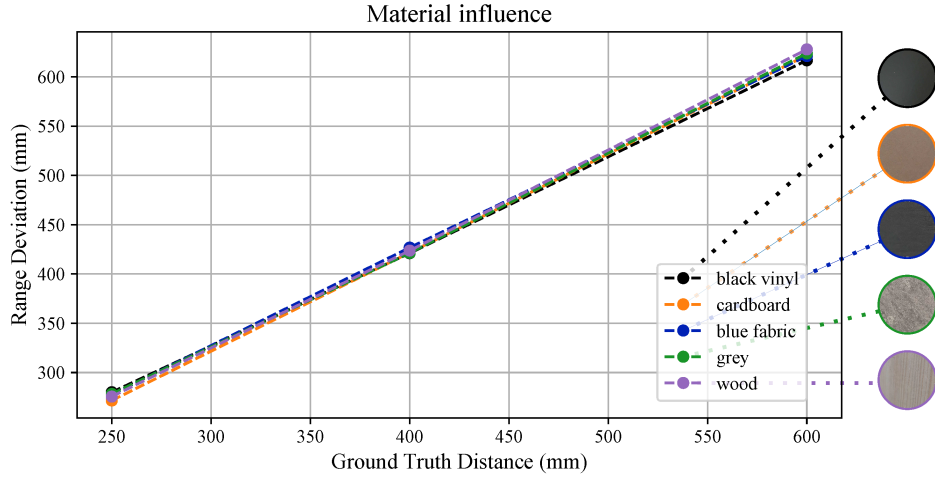


Figure 2.3.: Material properties influence on range estimation showing the estimated range at three distances for five separate materials. The figure is reused from the original paper [7].

2.1. Architectural Comparison

The physical and electrical trade-offs between different perception configurations and the proposed distributed sensing solution are summarized in Table 2.1.

Table 2.1.: Architectural Comparison of Sensing Schemes

Sensing Scheme	Data Output	Power & Weight Profile	Mounting & Coverage
Stereo Depth Camera (e.g., RealSense D435) [1]	Dense RGB-D/monochrome data (depth maps paired with color images)	Moderate power draw and moderate weight.	Rigid, bulky housing; typically restricted to a forward-facing view, leaving blind spots beneath and behind the robot.
3D LiDAR [2]	Sparse, very precise 3D point clouds	High power consumption and heavy weight.	Big cylindrical footprint; typically mounted on top of the chassis, leaving the underbelly unmonitored.
3D indirect ToF Camera (e.g., PMD Flexx2) [1]	Dense depth maps (distance encoded via phase shift)	Moderate power draw and low weight.	Compact compared to LiDAR, but still restricted to a single directional FoV (e.g., designated strictly for the underbelly).
Distributed Multi-Zone dToF Array (This Thesis)	Multi-Zone Ranging Matrices (8x8 grids)	Low power consumption (≈ 2 W total) and very low node weight.	Versatile; miniature sensor nodes can be scattered across multiple enclosures to provide coverage in many directions.

Chapter 3

Background

To provide a foundation for the multi-sensor architecture, this chapter establishes the mechanical, electronic, and mathematical baselines governing the proposed framework. The characteristics of the quadrupedal robot, the physics and layout of dToF sensors, and the MCU baseline are detailed.

3.1. The Quadrupedal Robotic Platform

The experimental platform for the sensing array is the Unitree Go1 quadrupedal robot, a legged platform engineered for agile navigation [1, 3]. The robot has a lightweight footprint with a total operational mass of approximately 12 kg, placing strict limits on any auxiliary payload configurations. Its kinematics are governed by twelve independent degrees of freedom, which are implemented through three active rotational joints per leg. Each joint is actuated by a custom high-torque brushless motor regulated via localized Field-Oriented Control drivers. Because the platform relies on a battery to maintain high-frequency joint control loops and primary telemetry processing, any added sensory hardware should preferably operate under tight power allocation boundaries. The placement of exteroceptive sensors is limited by the available space in the chassis. Therefore, the sensor system must be compact enough to fit where bulky LiDAR or stereo cameras cannot.

To manage these perception and estimation tasks efficiently, the platform splits the computational workload across a dual-compute architecture [1]. An NVIDIA Jetson Orin Nano is running the GPU-accelerated software pipelines, which include the NVIDIA Isaac ROS Visual SLAM for Visual-Inertial Odometry and the 2.5D local elevation mapping loops. The Extended Kalman Filter sensor fusion and high-level obstacle avoidance logic are handled by an Intel NUC featuring a 10th-generation i7 CPU. Within this framework, the MCU board developed in this thesis coordinates data collection by managing

3. Background

low-level communication across the sensor array. This layout manages sensor readings, establishing a dedicated pipeline to transmit processed data packets to the NUC and Jetson for future integration into the navigation loop.

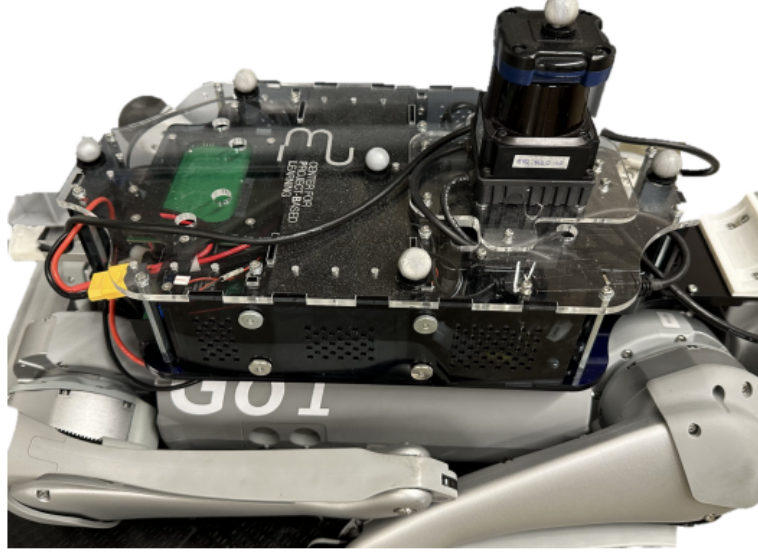


Figure 3.1.: Backpack V2 computing and power integration mounted to the Unitree Go1 robot chassis. The top-mounted LiDAR shown is unused for this thesis. Figure adapted from Schl pfer [3].

3.2. Microcontroller Baseline

Aggregating data from a six-sensor array without dropping frames requires a processing platform structured around hardware-accelerated peripherals. The STMicroelectronics STM32 portfolio provides a broad ecosystem categorized into tiers based on computational power, power efficiency, and hardware features, including Mainstream, Ultra-low power, Wireless, and High-performance lines. Mainstream or Ultra-low-power series handle simple control loops or low-duty-cycle tasks, though multi-zone proximity networks require continuous high-speed data acquisition, so the High-performance series is a necessity.

The ST High-performance portfolio includes MCU families designed to scale processing power for computationally demanding tasks. This tier features lines such as the STM32F7 and STM32H7 series running on ARM Cortex-M7 cores, the STM32H5 series with the Cortex-M33 core, and the specialized STM32N6 series operating on an advanced Cortex-M55 core [8, 9, 10, 11]. The clock frequencies in these families scale from 216 MHz to 800 MHz, providing the arithmetic capacity required to parse the raw distance matrices before transmitting data packets to robot computers.

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Beyond raw clock speeds, the key feature that justifies using the high-performance MCUs is that they offer extensive connectivity capabilities. They feature multiple independent communication buses, providing numerous channels for SPI, Inter-Integrated Circuit (I2C), UART, and USART (Universal Synchronous/Asynchronous Receiver/Transmitter) interfaces, while advanced variants within the portfolio also include CAN buses and a Flexible Memory Controller (FMC). This rich peripheral set allows multiple VL53L8CX sensors to be distributed across independent SPI channels, avoiding bus bottlenecks and reducing data acquisition latency.

3.3. Principles of Direct Time-of-Flight Sensing

Distance measurement within the proposed framework relies on the physics of dToF sensing. A dToF system determines the geometric range between the sensor face and a remote target surface by measuring the precise round-trip propagation time of a modulated light signal moving through the atmospheric medium. The total spatial distance d is derived using the constant speed of light c and the measured transit time interval Δt :

$$d = \frac{c \cdot \Delta t}{2}$$

Unlike indirect ToF architectures, which derive range implicitly by evaluating the continuous phase shift of sinusoidal waves and require extensive multi-frame calculations, dToF systems compute distance explicitly using discrete photon particle arrivals. The transmitter emits a high-energy nanosecond light pulse. As the emitted photons encounter an obstacle, a portion of the light scatters backward toward the sensor’s receiving aperture. The arrival of these returning photons is captured by an integrated matrix of high-sensitivity SPAD. Every registered single-photon impact triggers an instantaneous avalanche breakdown current within the semiconductor junction, which is captured by high-frequency Time-to-Digital Converters to lock the raw temporal duration Δt . This particle-counting methodology delivers distinct advantages for mobile robotics. First, it minimizes the raw current draw on the embedded battery system. Second, it provides native protection against optical saturation under bright ambient sunlight. Finally, it provides the mathematical flexibility needed to isolate multiple independent target reflections within a single optical FoV.

3.4. The VL53L8CX Architecture

The sensor deployed on the peripheral architecture is the VL53L8CX multi-zone ToF ranging module designed by STMicroelectronics [12]. The device integrates a 940 nm Vertical-Cavity Surface-Emitting Laser (VCSEL) emitter paired with a Diffractive Optical Element. This configuration shapes and projects a square $45^\circ \times 45^\circ$ light field to

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achieve a wide 65° diagonal system FoV, matching the optical constraints detailed in Figure 3.3. The underlying architecture shifts away from older single-pixel proximity sensing by dynamically segmenting its receiving SPAD matrix into distinct spatial zones [13]. The low-level device configuration dictates several fundamental operation principles:

- **Matrix Resolution Configuration:** The sensor firmware allows the host to configure the spatial output grid as either a high-speed 4×4 matrix (16 independent sensing zones) or a high-density 8×8 matrix (64 independent sensing zones) [13]. In the 8×8 configuration, the system sweeps an array of spatial vectors across the environment, tracking independent targets simultaneously up to a maximum physical ranging limit of 400 cm [12].
- **On-Chip Histogram Processing:** Every individual zone independently tracks incoming photon reflections over time, accumulating the data inside a local discrete temporal histogram. An onboard MCU core runs real-time digital signal processing algorithms directly on these histogram shapes to distinguish true target peaks from ambient noise. Further, it calculates distances and determines confidence levels per zone [13].
- **Multi-Target Detection:** Because the sensor’s on-chip firmware analyzes internal photon distribution histograms rather than relying on basic amplitude threshold crossings, each matrix zone can isolate and report up to four distinct target distances within its specific light cone. If a thin obstacle or glass pane partially obstructs a zone, the sensor can register the close obstacle distance while simultaneously capturing the secondary background floor reflection in the same data frame.
- **Crosstalk and Motion Tracking:** The module features hardware-accelerated compensation blocks to filter out veiling glare and internal multi-path optical crosstalk induced by protective cover plates. Additionally, the sensor executes a spatial motion indicator algorithm across successive frames, appending motion vector scalars to each zone’s data packet to flag dynamic objects.
- **Framing:** The VL53L8CX can stream continuous ranging matrices at frequencies up to 60 Hz in 4×4 mode, or up to 15 Hz when running full 8×8 resolution sweeps [13]. It communicates over high-speed digital buses, utilizing an SPI interface running at up to 3 MHz or an I2C running at up to 1 MHz to ensure real-time data streaming capabilities.

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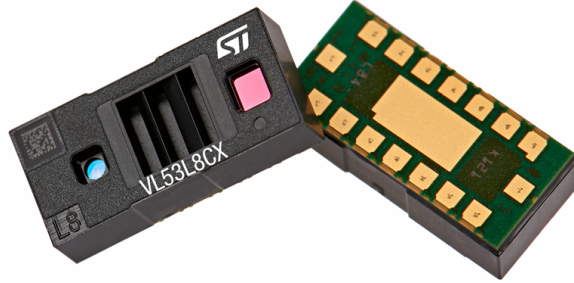


Figure 3.2.: The package configuration of the VL53L8CX dToF sensor, detailing the transmitter optical lens and receiver SPAD arrays. Figure adapted from the VL53L8CX datasheet [12].

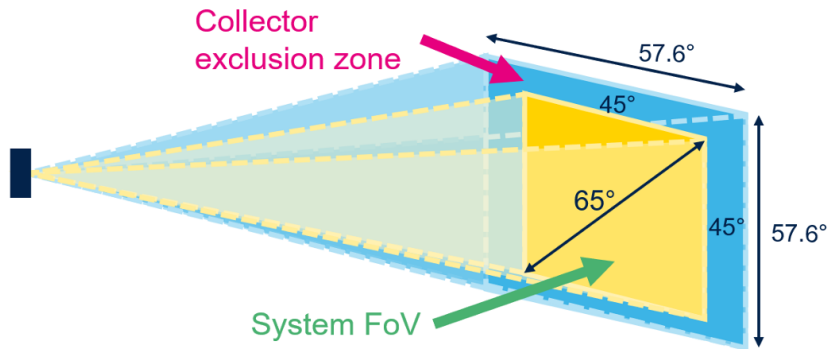


Figure 3.3.: Geometric definition of the system FoV cone compared against the raw collector exclusion zone boundaries. The yellow region maps out the true $45^\circ \times 45^\circ$ (65° diagonal) sensing cone, while the outer blue perimeter defines the $57.6^\circ \times 57.6^\circ$ clearance boundary necessary to prevent false target returns from reflective enclosure walls. Figure adapted from the VL53L8CX datasheet [12].

3.5. Emerging Horizons: The VL53L9CX Module

The scalability of distributed proximity arrays is tied to rapid advances in semiconductor rangefinder technology. A significant milestone in this domain is the newly released VL53L9CX 3D dToF lidar module, representing the next step in STMicroelectronics' multi-zone product lifecycle [14]. While the VL53L8CX provides an excellent baseline with its sparse 8×8 spatial layout, the new VL53L9CX advances the matrix-based tracking paradigm into high-resolution LiDAR territory. VL53L9CX significantly scales the spatial density, supporting massive multizone configurations up to a dense matrix grid 54×42 of separate tracking zones. This enhanced resolution is powered by dual 940 nm VCSEL illumination lasers coupled with Metasurface Optical Elements, generating a 71° diagonal field of view. The module extends the maximum tracking range from 4 meters up to 8.8 meters. Additionally, its on-chip histogram processing loops allow it to stream 2D infrared images, dense depth maps, and ambient light profiles at frame rates up to 100 Hz. Although hardware development in this thesis focuses on the VL53L8CX framework, shared firmware philosophies and dToF technology of the VL53L9CX represent an exciting future path. The multi-sensor methodologies developed in this work can scale up to drive dense peripheral arrays, though managing multiple concurrent streams will increase the processing and communication load on the robot's main computer.

3.6. Computer-Aided Design

The implementation of the sensor peripheral network required a hardware solution developed within the open-source KiCad Electronic Design Automation software suite. KiCad was utilized for schematic design and multi-layer PCB routing. The board layout was developed according to best practices in hardware design, optimizing signal integrity, eliminating potential ground loops, and mitigating crosstalk.

Chapter 4

Hardware Design and Firmware Implementation

This chapter details the design and implementation of the hardware and firmware. It outlines the system architecture, schematic components, PCB layout, and sensor configurations.

4.1. System Architecture Overview

The distributed proximity sensing network is centered on a high-speed master PCB (the MCU board) and six independent nodes (the sensor boards). The six sensor boards are grouped into two groups. Each group is allocated to one independent SPI channel provided by the STM32F723VET6 MCU. Thus, ensuring that data retrieval blocks run in parallel without bus bottlenecks [8, 12]. Bandwidth calculations verified that putting three and even four sensors on a single SPI bus is reasonable, and implementation confirmed that the system operates well under three sensors per bus configuration. To handle the resulting high-throughput payload, the MCU utilizes its integrated on-chip USB HS interface to stream the proximity matrices to the robot computer.

4.1.1. Module Connectivity

The interface between the MCU board and the sensor boards is routed through low-profile Hirose FH34SRJ connectors [15]. Because this connector is a ZIF, it provides excellent mechanical stability and saves valuable vertical space on the board; however, the primary reason for choosing it was its dual-contact feature, which guarantees contact reliability regardless of cable insertion orientation. To bridge the sensor boards to the

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MCU core, the interconnect utilizes Flexible Flat Cable (FFC) rather than conventional wire bundles. Traditional multi-wire bundles introduce physical bulk, add dead weight, and create inconsistent twisting geometries that produce parasitic capacitance and open lines to Electromagnetic Interference (EMI). Conversely, standard FFC enforces a uniform layout that isolates high-frequency SPI signal transitions and distributes mechanical strain evenly under the vibration locomotion profile.

During the design phase, alternative cabling possibilities were considered. Flexible Printed Circuit (Flexible Printed Circuit (FPC)) cables essentially give all the advantages of FFC cables, but they are specifically needed if the design requires continuous movement (like in robotic joints), extreme heat resistance (FPCs easily withstand temperatures over 200°C, whereas many standard FFCs are limited to around 80°C to 105°C), or advanced routing across multilayer stacks; i.e., in very harsh environments. Because the sensor boards will remain mostly static and are bent only once during installation, FPC cables are an unnecessary alternative that is five times more expensive. Micro-Coaxial Cables were also considered since they are robust and also do not take up vertical space, and are (most) importantly shielded; however, at more than 10 CHF for a single bundle, they proved too expensive for this application. Consequently, FFCs (0.5 mm pitch, 10 Contacts, A Type, L=152mm, Gold) were selected as the optimal choice.

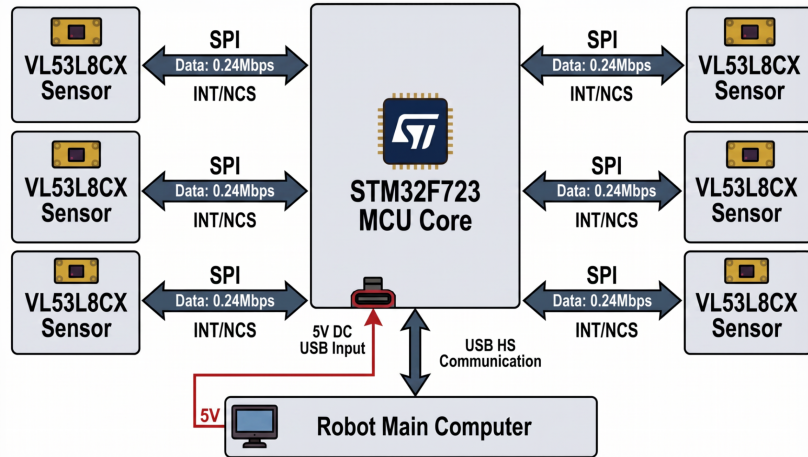


Figure 4.1.: Diagram illustrating the connections between the central MCU board and the six external sensor boards over the two independent SPI channels

4.2. Schematic Design and Component Selection

4.2.1. Microcontroller Selection

The MCU for the motherboard is chosen by considering specific electrical and computational parameters. A primary differentiator during the hardware design stage was

4. Hardware Design and Firmware Implementation

the architecture of the integrated USB peripheral. Standard high-performance MCUs typically feature an on-chip USB Full-Speed (FS) operating at a maximum throughput of 12 Mbps [16]. To achieve USB HS throughput 480 Mbps, conventional designs must implement an external physical-layer (PHY) transceiver chip, such as the Microchip USB3300, interfaced via a UTMI+ Low Pin Interface (ULPI) bus. The protocol is named a *Low Pin Interface* relative to older parallel layout standards, but it still demands 12 dedicated, high-frequency digital signal tracks running at 60 MHz. This layout complicates board design (track length matching requirements) and consumes significant PCB space. Finally, adding an external transceiver integrated circuit and its associated passives increases the component count, creating extra potential points of failure.

Selecting an MCU with an on-chip USB HS PHY reproduces this entire layout onto the silicon die. This design choice eliminates the ULPI tracking interface, saves critical PCB area, and guarantees a reliable upstream interface [16, 17]. The performance jump from a USB FS link to a USB HS link is fundamental for the scalability and operational safety of the platform. At maximum capabilities, a single multi-zone sensor data frame can consume roughly 2,000 bytes of data. For our six-sensor array running in the high-density 8×8 matrix configuration at its maximum frame rate of 15 Hz, the raw data throughput amounts to:

$$6 \text{ sensors} \times 2,000 \text{ bytes} \times 15 \text{ Hz} = 180,000 \text{ bytes per second} \approx 1.44 \text{ Mbps}$$

of raw bandwidth. While this baseline throughput fits within the 12 Mbps ceiling of a standard USB FS link, real-world FS efficiency is degraded by USB protocol packet overhead, framing tokens, and multi-device bus arbitration.

Crucially, a USB FS connection can introduce data bottlenecks and transmission delays under multi-sensor loads. In robotics, any latency or delayed packet delivery translates to dangerous blind spots in the obstacle avoidance loop. For a quadruped, lagging data means the navigation computer receives obstacle vectors late, which can result in collisions with the environment. Moving to a 480 Mbps HS interface provides a 40-fold increase in bus capacity. This minimizes bus-induced latency and guarantees a future-proof communication pipe capable of streaming high-frequency proximity data from 10 or more sensors without the risk of packet lag.

Beyond the USB sub-system, several hardware parameters restricted the selection pool:

- **Package:** The chip must be a Low-Profile Quad Flat Package (LQFP) to allow for hand-soldering and hardware debugging within a standard laboratory environment. Ball Grid Array (BGA) was excluded due to its reliance on advanced reflow ovens and X-ray inspection.
- **Pin Count:** A 100-pin is the optimal footprint. Smaller 64-pin variants are generally absent from high-performance MCU lines and offer insufficient General-Purpose Input/Output (GPIO) pins to route the dedicated interrupt (INT) and not chip select (NCS) lines needed for the SPI protocol for each sensor. Conversely, 144-pin or

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176-pin variations increase the footprint beyond the compact limits, though they are acceptable.

- **Memory and Clock Dynamics:** The core must operate above 200 MHz to maintain deterministic control loops, backed by a of 512 KB of Flash memory and 216 KB of Static Random-Access Memory (SRAM) to buffer incoming multi-zone target matrices. These memory baselines are necessary because the vendor-supplied Ultra Lite Driver (ULD) API libraries carry a large code footprint. Managing distance grids requires large runtime buffer allocations to prevent stack overflows or memory exhaustion.
- **Connectivity:** The application requires a single-core line with at least three SPI channels. Dual-core processing lines and dedicated hardware cryptography accelerators were avoided to eliminate firmware complexity and prevent unnecessary power overhead. Specialized hardware blocks, such as advanced mathematical vector accelerators, neural network engines, DSP extensions, or dedicated graphics pipelines, were not a priority for this implementation, as the objective is strict serial data routing and packet formatting rather than heavy computing tasks.

Four high-performance candidates were assessed using the table shown in Figure 4.2.

Feature ⁽¹⁾	STM32F401 STM32F411		STM32F415 STM32F407 STM32F417 STM32F427 STM32F437 STM32F429 STM32F439		STM32F412 STM32F413 STM32F423		STM32F446 STM32F469 STM32F479 STM32F74x STM32F756 STM32F76x STM32F77x STM32F7x2 STM32F730R/V STM32F750		STM32F7x3 STM32F730Z/I		STM32H742 STM32H743 STM32H753 STM32H750 STM32H745 STM32H755 STM32H747 STM32H757		STM32H7A3 STM32H7B3 STM32H7B0 STM32H723 STM32H733 STM32H735 STM32H730		STM32H5F/E STM32H7R3x8 /R7x8 STM32H7S3x8/ S7x8 STM32H7S3x8/ S7x8 STM32H7R3x8 /R7x8		STM32C562 STM32H503 STM32H563 STM32H573 STM32H562		STM32N6
	OTG_FS	OTG_HS	OTG_FS	OTG_HS	OTG_FS	OTG_HS	OTG_FS	OTG_HS	OTG_FS	OTG_HS	OTG_FS	OTG_HS	OTG_HS	OTG_FS	OTG_HS	OTG_FS	OTG_HS	USB 2.0 FS host/device interface	
Crystal-less USB	-	-	-	-	-	-	-	-	-	-	X	X	X	X	X	X	X	-	-
Bidirectional endpoints (including EP0)	4	6	4	6	6	9	6	9	6	9	6	9	9	9	6	9	8	8	9
Host mode channels	8	12	8	12	12	16	12	16	12	16	12	16	16	16	12	16	8 ⁽²⁾	8	16
Size of dedicated packet buffer SRAM (Kbytes) ⁽³⁾	1.25	-	1.25	4	1.25	-	1.25	4	1.25	4	1.25	4	4	4	1.25	4	2	2	4
Integrated PHY	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	HS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	HS ⁽⁴⁾	FS ⁽⁴⁾	FS ⁽⁴⁾	HS ⁽⁴⁾
ULPI available to primary I/Os via multiplexing	-	-	-	X	-	-	-	X	-	-	-	-	-	-	-	-	-	-	-
Dedicated VDDUSB	NO	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
LPM	-	-	-	-	X	-	X	X	X	X	X	X	X	X	X	X	X	X	X
BCD	-	-	-	-	X	-	-	-	X	-	X	X	X	X	X	X	X	X	X
ADP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Figure 4.2.: Table illustrating USB implementation on high-performance products. Red circles highlight the on-chip HS PHY. Figure adapted from AN4879 [16]

4. Hardware Design and Firmware Implementation

The evaluation targeted the following models across high-performance generations:

1. **STM32H5E5ZJ**: A highly efficient chip running at 250 MHz packaged in a 144-pin LQFP, featuring an integrated USB HS PHY alongside an 4 MB of Flash and 1536 KB of SRAM [10]. Although highly capable, this series was brand new during the development cycle, with its primary software ecosystem and production hardware rolling out between February and March 2026. Consequently, it was completely unavailable via standard distribution channels like Mouser or DigiKey. Furthermore, it provides way more memory than needed for the thesis applications.
2. **STM32H723VG6**: A very well documented (including many tutorials and articles about it on the Internet) high-performance component operating at 240 MHz in a compact 100-pin LQFP, backed by 1 MB of Flash and 564 KB of SRAM [9]. Despite its excellent computational benchmarks, this variant only incorporates an on-chip FS transceiver, requiring an external transceiver over a ULPI routing layout to realize high-speed operation [9, 16].
3. **STM32N645Z0**: An advanced dual-core module operating at 800 MHz with 4 MB of Flash and 4.2 MB of internal memory and an on-chip USB HS PHY [11]. Further, it has a dedicated neural network engine, a computer vision pipeline, and an accelerator for graphics. This processing capability vastly exceeds the algorithmic tracking requirements of a proximity array. Finally, it is exclusively offered in ultra-fine-pitch VFBGA packages, making manual assembly and lab debugging very hard.
4. **STM32F723VET6**: The selected component. It perfectly satisfies every structural and electrical constraint, offering an integrated on-chip USB HS PHY inside a hand-solderable 100-pin LQFP footprint [8]. It operates at 216 MHz with an allocation of 512 KB of Flash, 216 KB of SRAM, and four SPI channels without the layout overhead of unused cryptography hardware blocks [8].

A summary of this architectural evaluation is presented in Table 4.1.

Table 4.1.: High-Performance MCUs Comparison

MCU Model	Package	Freq MHz	Flash/RAM	HS PHY	Selection Status
STM32H5E5ZJ	LQFP144	250	4 MB/1536 KB	On-Chip	Unavailable
STM32H723VG	LQFP100	240	1 MB/564 KB	External	Rejected (ULPI)
STM32N645Z0	VFBGA144	800	4 MB/4.2 MB	On-Chip	Rejected (BGA)
STM32F723VE	LQFP100	216	512 KB/216 KB	On-Chip	Selected

4.2.2. USB Hardware Protection and Clocking Implementation

Operating a high-speed data link at 480 Mbps requires dedicated hardware protection and a stable clock source. Joint commutation and motor drivers plus the USB HS

4. Hardware Design and Firmware Implementation

itself generate significant electromagnetic interference and transient voltage surges across shared grounding paths, exposing sensitive differential communication lines ($D+$ and $D-$) to electrostatic discharge (Electrostatic Discharge (ESD)) hazards.

To counter these surges without distorting high-speed communication signals, a protection circuit was implemented following official reference layouts [16]. The protective topology is illustrated in Figure 4.3.

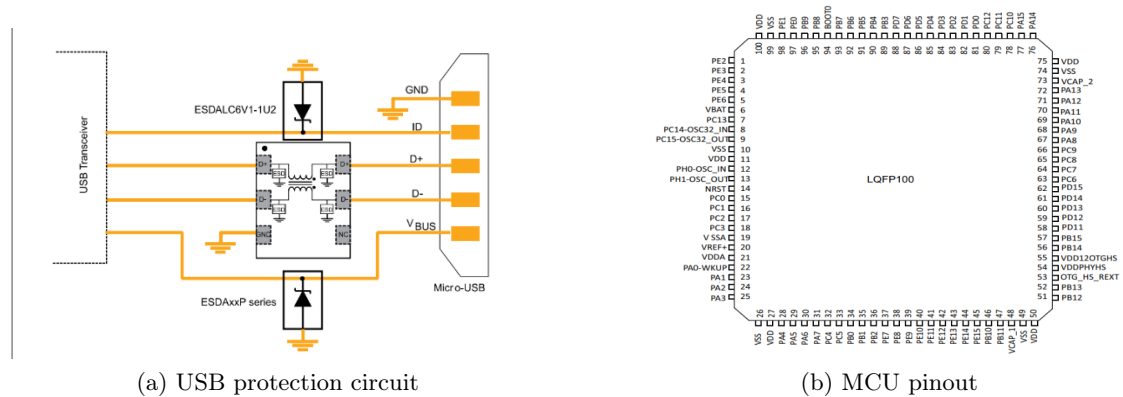


Figure 4.3.: MCU pinout and its corresponding USB high-speed protection and filtering circuit. Figures adopted from [16] and [8]

The physical layout incorporates two protection components placed at the entrance of the USB interface connector:

- **STMicroelectronics ECMF02-2AMX6:** A common-mode filter featuring integrated bidirectional ESD protection diodes [18]. High-speed differential routing requires minimal line loading to prevent rise-time degradation and data attenuation. The ECMF02-2AMX6 provides a wide differential bandwidth exceeding 3.5 GHz and exhibits a low parasitic capacitance of only 0.6 pF [18]. This allows it to suppress high-frequency noise and sudden ESD spikes without degrading the integrity of the 480 Mbps signals.
- **STMicroelectronics ESDA7P60:** A unidirectional Transient Voltage Suppressor (TVS) diode assigned to the V_{BUS} line [19]. This diode safely suppresses high-energy inductive voltage surges and transients up to 70 A (8/20 μ s pulse), preventing overvoltage damage from propagating into the downstream 3.3 V and 1.8 V DCDC converters [19].

Similarly, the internal clock system must be stabilized because the USB HS peripheral cannot function without a stable clock reference[20]. The internal oscillators are sufficient for basic processing, but the 480 Mbps High-Speed protocol demands an absolute timing precision that internal hardware cannot supply; the internal oscillators cannot maintain a precise reference frequency. To prevent data corruption and bus drops, a high-precision

4. Hardware Design and Firmware Implementation

external crystal oscillator (HSE) is mandatory to drive the internal phase-locked loops (PLL) and lock the USB clock domain safely [21, 17]

An Abracon ABM8W-25.0000MHZ-8-D1X-T3 quartz crystal operating at a resonance frequency of 25 MHz is used for that purpose[22]. This component is specifically designed for Internet of Things (IoT) hardware and machine-to-machine (M2M) links featuring a tight frequency tolerance of ± 10 ppm and good aging characteristics [22]. Using this crystal, the hardware configuration provides a thermally immune reference clock [21].

4.2.3. Debugging and Programming Interface

For system configuration, the MCU board incorporates a dedicated Serial Wire Debug (SWD) port [23]. The interface is implemented using a Samtec FTSH-105-01-L-DV-K-A connector, which provides a standard 10-pin ARM Cortex Debug footprint. Flash programming is performed using a Segger J-Link debugger. The programming port links to dedicated pins on the STM32F723VET6 to drive the debug infrastructure [8]. Specifically, the bidirectional data line (SWDIO) is mapped to pin PA13, the clock line (SWCLK) to pin PA14, and the asynchronous trace output line (SWO) to pin PB3. The hardware reset (SWD_NRST) is routed to the reset pin of the MCU.

To maintain signal integrity across these lines, a series of $33\ \Omega$ resistors are placed between the J7 connector pins and the MCU ports [24, 20, 25]. They are positioned as close to the SWD connector and serve a dual purpose: they function as current-limiting elements to shield the MCU and act as dampers, maintaining error-free debugging.

4.2.4. Power Infrastructure and Current & Power Calculations

Establishing a robust power supply is vital for isolating the sensor array from the voltage drops and electrical noise generated by a robot's actuators. The MCU board operates via a 5.0 V DC input sourced from an upstream computer interface. This input voltage must be stepped down to a clean 3.3 V to drive the MCU core and the sensor boards.

During the initial design, an LD39050PU33R ultra-low-dropout (LDO) linear regulator was considered for the 5.0 V to 3.3 V conversion. However, current estimations forced its abandonment; it provides 500 mA maximum current, which theoretically would be enough, but practically the component would operate at its limit, producing a lot of heat and eventually damaging its operation. Linear regulators exhibit poor efficiency, calculated as the ratio of output voltage to input voltage ($\eta \approx V_{\text{out}}/V_{\text{in}} = 3.3/5.0 = 66\%$). The remaining 34% of the power consumption would be dissipated as heat.

To maximize power efficiency and mitigate thermal buildup, a Texas Instruments buck converter TPSM84203 was chosen [26]. It has efficiency up to 95% and provides an integrated solution. It combines a 1.5 A DC/DC converter with matching power MOSFETs, a shielded inductor, and compensation passives into a compact 3-pin through-hole

4. Hardware Design and Firmware Implementation

package. This integration simplifies the layout by eliminating the need for selection of magnetic components. The module requires a minimum input capacitance of $10\ \mu\text{F}$ and a minimum output capacitance of $94\ \mu\text{F}$, consisting of two $47\ \mu\text{F}$ capacitors to ensure low output ripple voltage. High-quality X5R or X7R ceramic capacitors in a 1210 package were selected to fulfill these criteria, satisfying the vendor's voltage rating recommendations.

To validate this power architecture, independent current and power calculations were derived using operational baselines from the component datasheets. The current profile of the system is structured as follows:

1. **Sensor Load:** Each VL53L8CX module draws a typical active ranging current of 50 mA [12]. For the full array, the sensor current is 300 mA
2. **MCU Consumption:** Firmware execution on the STM32F723VET6 operates at a core clock frequency of 216 MHz directly out of Flash memory via the AXI interface, with the internal regulator turned on, the L1-cache activated, and the ART Accelerator enabled [8]. The datasheet defines a current baseline of 82 mA with all hardware peripherals disabled, rising to 155 mA when all are enabled [8]. Because the proximity code disables the majority of peripherals, the active core operational current consumption settles at 95 mA [8].
3. **LEDs Currents:** There are a 5.0 V monitor LED (D2) and a 3.3 V monitor LED (D3) on the MCU board, alongside a 3.3 V power indicator LED (D1) on each of the six sensor boards. All status indicators utilize green LEDs with a forward voltage of $V_f = 2.85\ \text{V}$. Current consumption through these indicators and their respective current-limiting resistors is derived:

$$I_{\text{LED1}, 5\text{V}} = \frac{5.0\ \text{V} - 2.85\ \text{V}}{480\ \Omega} = 4.48\ \text{mA} \quad (4.1)$$

$$I_{\text{LED2}, 3.3\text{V}} = \frac{3.3\ \text{V} - 2.85\ \text{V}}{480\ \Omega} = 0.94\ \text{mA} \quad (4.2)$$

$$I_{\text{LED, sensor}} = \frac{3.3\ \text{V} - 2.85\ \text{V}}{680\ \Omega} = 0.66\ \text{mA} \implies I_{\text{LED, sens_total}} = 6 \times 0.66\ \text{mA} = 3.97\ \text{mA} \quad (4.3)$$

Summing these loads and accounting for all passive components (like pull-up resistors) on the boards, the current consumption for the system is:

$$I_{\text{total}} = 300\ \text{mA} + 95\ \text{mA} + 4.48\ \text{mA} + 0.94\ \text{mA} + 3.97\ \text{mA} = 404.39\ \text{mA} \approx 405\ \text{mA} \quad (4.4)$$

The power profile of the system is structured as follows: Continuous ranging in an 8×8 zone demands a power of 215 mW per sensor [12]. Across six nodes, the total power is 1290 mW. The power consumed by the MCU operating at 3.3 V is:

$$P_{\text{MCU}} = 3.3\ \text{V} \times 95\ \text{mA} = 313.5\ \text{mW} \quad (4.5)$$

4. Hardware Design and Firmware Implementation

Each sensor board integrates a DC/DC to step down the incoming 3.3 V to the 1.8 V required by the ToF sensor [27, 12]. Under an active 50 mA load, the local sensor regulator operates at an efficiency of $\eta = 93\%$ [27]. The output power ($P_{\text{out, sens}}$), input power pulled from the motherboard ($P_{\text{in, sens}}$), and component power loss ($P_{\text{loss, sens}}$) per sensor board are derived as:

$$P_{\text{out, sens}} = V_{\text{out}} \times I_{\text{out}} = 1.8 \text{ V} \times 0.050 \text{ A} = 0.090 \text{ W} = 90 \text{ mW} \quad (4.6)$$

$$P_{\text{in, sens}} = \frac{P_{\text{out, sens}}}{\eta} = \frac{90 \text{ mW}}{0.93} = 96.77 \text{ mW} \quad (4.7)$$

$$P_{\text{loss, sens}} = P_{\text{in, sens}} - P_{\text{out, sens}} = 96.77 \text{ mW} - 90 \text{ mW} = 6.77 \text{ mW} \quad (4.8)$$

Total power loss attributed to voltage translation in sensor boards:

$$P_{\text{loss, sens_total}} = 6 \times 6.77 \text{ mW} = 40.62 \text{ mW} \quad (4.9)$$

The power introduced by the passive current-limiting resistors and LEDs across the system is:

$$P_{\text{indicating_total}} = (5.0 \text{ V} \times I_{\text{LED1}}) + (3.3 \text{ V} \times I_{\text{LED2}}) + 6 \times (3.3 \text{ V} \times I_{\text{LED, sensor}}) \quad (4.10)$$

$$P_{\text{indicating_total}} = 22.40 \text{ mW} + 3.10 \text{ mW} + 13.11 \text{ mW} = 38.61 \text{ mW} \quad (4.11)$$

Finally, operating at an efficiency of 95% ($\eta_{\text{mcu}} = 0.95$) under the calculated current, the converter's losses are derived using the same method (with $V_{\text{out}} = 3.3 \text{ V}$ and $I_{\text{out}} = 405 \text{ mA}$) for the sensor nodes [26]. This yields a power loss of $P_{\text{mcu_dcdc_loss}} = 70.23 \text{ mW}$. Summing all power domains and accounting for all passive components on the boards yields:

$$P_{\text{total}} = P_{\text{sensors_active}} + P_{\text{MCU}} + P_{\text{indicating_total}} + P_{\text{loss, sens_total}} + P_{\text{mcu_dcdc_loss}} + P_{\text{passives}} \quad (4.12)$$

$$P_{\text{total}} = 1290 \text{ mW} + 313.5 \text{ mW} + 38.61 \text{ mW} + 40.62 \text{ mW} + 70.23 \text{ mW} = 1753 \text{ mW} \approx 1.80 \text{ W} \quad (4.13)$$

To verify the accuracy of this result, the total current is multiplied by the system's input voltage:

$$P_{\text{check}} = I_{\text{total}} \times V_{\text{in}} = 405 \text{ mA} \times 5.0 \text{ V} = 2.025 \text{ W} \quad (4.14)$$

The two results match within an acceptable error window, proving that the module operates at 2 W and fits safely within the current rating of the TPSM84203 [26].

4.2.5. Sensor Board Design

The sensor board receives 3.3 V from the MCU board. To power the VL53L8CX sensor, a Texas Instruments TPS62840 buck converter steps this voltage down to 1.8 V [27]. The

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MCU communicates using 3.3 V logic while the sensor operates on a 1.8 V domain. Thus, a voltage level translation is mandatory across all signal tracks. Standard auto-direction level translators can struggle with edge distortion and propagation delays. To prevent this, unidirectional translators are implemented to keep the digital signal clean.

To handle signals sent from the MCU to the sensor node, a four-channel fixed-direction voltage translator TXU0104 is implemented [28]. This IC steps down the 3.3 V master signals to the 1.8 V, routing the Master Output Slave Input (MOSI), Master Clock (MCLK), No-Chip Select (NCS) and Synchronization (SYNC) lines.

Conversely, the signals from the sensor to the MCU must be stepped up from 1.8 V to 3.3 V. Rather than utilizing a single multi-channel translator like a TXU0102 for the two output lines, this layout splits the signals using two single-channel TXU0101 voltage translators [29]. Splitting the Master Input Slave Output (MISO) and Interrupt (INT) paths minimizes cross-talk and provides control over the individual data paths.

An interesting detail of this architecture is the high-impedance state circuit. The TXU0101 (assigned to the MISO path) is controlled by a RUM001LD2TZCL N-channel MOSFET tied to (NCS), as shown in Figure 4.4b. When the MCU brings the NCS line high to deselect this specific sensor board, the change activates the gate of the MOSFET. This ties the Output Enable (OE) pin of the translator to ground, forcing its output state into a high-impedance mode. This disconnects the inactive sensor node from the shared data lines, preventing bus clashing and allowing multiple sensors to operate on the same SPI channel.

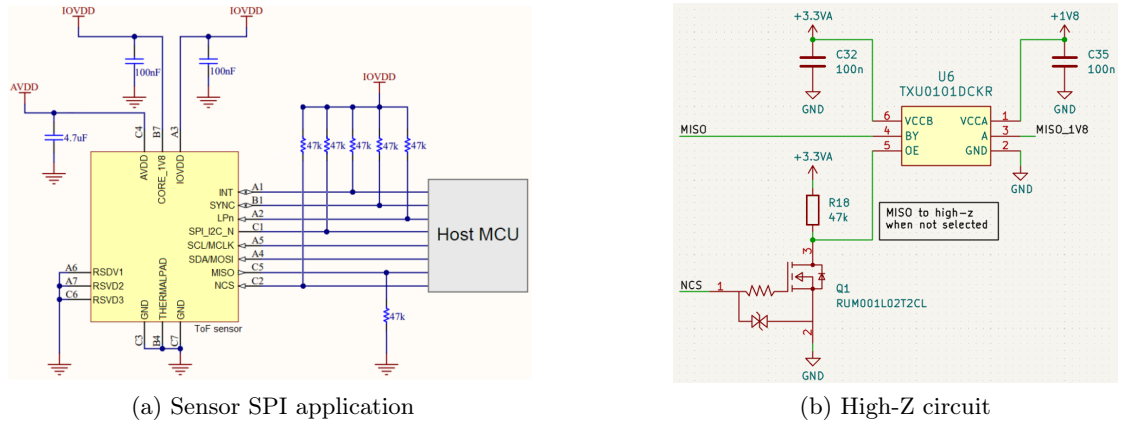


Figure 4.4.: SPI schematic and isolation circuitry for the voltage translator

Finally, the passive components stabilize the signal lines during active ranging. In accordance with the VL53L8CX reference design illustrated in Figure 4.4a, 47 k Ω pull-up resistors are added across data tracking pads [12]. These resistors prevent signal floating and dropouts inside the noisy environment of the robot.

4.3. Layout Design

To ensure signal integrity and minimize EMI, a strict PCB layout was executed. Both the MCU board and the sensor boards utilize an identical four-layer stackup configuration, via-stitching methodology, and solder mask constraints to match environmental tolerances.

The four-layer stackup is ordered as: Layer 1 (Top Signal/GND polygon) – Layer 2 (Ground Plane) – Layer 3 (Power Plane) – Layer 4 (Bottom Signal/GND). To suppress EMI radiating from SPI and USB data lines, continuous ground copper polygons are poured across Layers 1, 2, and 4. Layer 3 is dedicated to a solid 3.3 V power plane. This configuration creates a layout pairing between the planes, establishing low-impedance return paths underneath signal traces and reducing the board’s loop inductance. The boards are manufactured with a solder mask expansion of 0.08 mm and a minimum mask web width of 0.1 mm, ensuring copper pad isolation and eliminating the risk of solder bridging during assembly.

Many stitched ground vias are placed across the copper to tie the reference planes together. This stitching prevents the formation of ground loops that could distort the sensor measurements.

The component placement is kept as tight as possible to achieve a clean board. Most surface-mount components have a 0603 footprint to preserve area, whereas bulk capacitors are assigned to larger 0805 packages to handle higher ripple currents. Strict routing rules govern the layout:

- The 25 MHz crystal oscillator is placed far away from the DC/DC and the USB-C to protect the clock from noise jitter.
- The USB-C connector and the DC/DC are grouped closely to contain high-current loops and isolate supply spikes at the board entry.
- Ceramic decoupling capacitors are positioned as close as possible to their respective V_{DD} supply pins on both the MCU and the dToF sensors, filtering high-frequency noise.

Standard tracks and high-speed data lines are routed using a trace width of 0.2 mm. Power delivery traces are widened to 0.5 mm to minimize trace resistance and limit trace heating.

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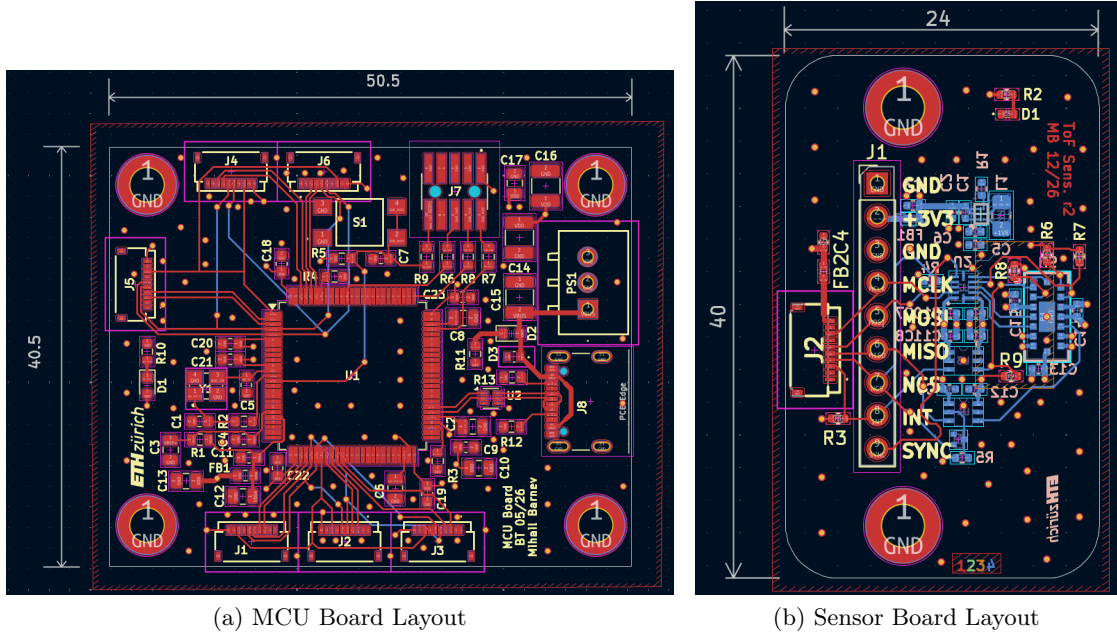


Figure 4.5.: Layout in KiCad for the circuits and their size.

4.4. Sensor Firmware Configuration

The firmware drives the distributed sensor topology, packaging proximity matrices for upstream transmission. The software architecture was developed using the STM32CubeIDE development environment, leveraging STM32CubeMX to initialize clock trees and pin configurations. The firmware integrates the STMicroelectronics API libraries alongside a USB Device Class stack configured for Communication Device Class virtual com port emulation [30].

Upon system boot, the firmware first checks if the sensor is responding by calling `vl53l8cx_is_alive()`. Once physical bus communication is verified, the firmware loads configuration parameters, initializes the sensor structures, and launches the active measurement sequence using `vl53l8cx_start_ranging()`. Inside the main application loop, the MCU continuously retrieves the fresh distance matrices using `vl53l8cx_get_ranging_data()` [13].

4.4.1. Operational Regimes and Parameter Optimization

Optimizing sensor network parameters balances tracking accuracy against processing and data latency. The operational settings selected for the VL53L8CX are combined in Table 4.2 [13].

4. Hardware Design and Firmware Implementation

Table 4.2.: Ranging Regimes and Parameter Explanation

Parameter	State	Explanation
Matrix Resolution	8×8	Configures 64 discrete zones for high-density terrain mapping.
Ranging Freq.	15 Hz	Set to optimize the photon integration window per zone.
Ranging Mode	Continuous	Keeps the VCSEL active during all ranging to maximize distance and ambient immunity.
Target Order	Closest	Priorities the nearest object reflections to eliminate imminent collision risks.
Targets Per Zone	2 Targets	Captures thin, closer obstructions without losing tracking of the high-reflectance background targets.
Sharpeners	20%	Effectively eliminates veiling glare and cross-zone distortion without inducing artificial target shrinkage.
VHV Calibration	450 (Every 30 s)	Triggers periodic temperature compensation to eliminate tracking offset drift induced by sensor heating.
SYNC Pin	Unused	External synchronization features are only available in Autonomous mode.

The rationale behind these selections balances the following trade-offs:

- **Resolution vs. Frequency:** While a 4×4 grid supports faster rates up to 60 Hz, it lacks the density needed to map narrow terrain changes. Conversely, an 8×8 grid can be pushed up only to 20 Hz in ideal test environments. However, limiting the rate to 15 Hz (as suggested by STM) extends the photon integration window. This extended window ensures that each zone receives sufficient backscattered light to track low-reflectance obstacles.
- **Continuous Mode Selection:** The Continuous ranging forces the device to continuously grab frames at the defined frequency with the VCSEL enabled during all ranging. Compared to the Autonomous mode, which limits the laser illumination window to reduce power draw, keeping the VCSEL always active allows for applications where fast ranging measurements are required.
- **Target Sorting Dynamics:** The default `TARGET_ORDER_STRONGEST` setting returns the object with the highest optical reflection peak. This is stable in clear space, but it can cause the sensor to miss a thin, low-reflectance object (like a dark table leg) standing in front of a bigger, highly reflective surface (like a white wall). Utilizing `TARGET_ORDER_CLOSEST` forces the sensor to report the shortest distance first, ensuring that semi-transparent or narrow obstructions are captured in the navigation loop.
- **Advanced Plugin Deactivation:** Optional software features (such as integrated motion indicators, software detection thresholds, and interrupt autostop mechanics)

4. Hardware Design and Firmware Implementation

were omitted from the firmware loop. These processing plugins require compiled code additions that exhaust the SRAM. Furthermore, because the quadrupedal chassis is constantly moving during locomotion, the entire environment appears to move relative to the observer, making motion-tracking algorithms redundant.

4.4.2. Output Selection Profile

To minimize data processing overhead on the MCU, non-essential data domains were deactivated within the driver configuration registers. Table 4.3 isolates the retained tracking variables[13].

Table 4.3.: Sensor Output Configuration

Output Param.	Status	Function
DISTANCE_MM	ENABLED	Provides the millimeter-accurate depth essential for obstacle avoidance.
TARGET_STATUS	ENABLED	Critical data quality byte used to check valid measurements.
NB_TARGET_DETECTED	ENABLED	Tracks multi-target reflections within each zone's FoV cone.
SIGNAL_PER_SPAD	ENABLED	Monitors signal return health to detect sensor blindness or dirty optical surfaces.
RANGE_SIGMA_MM	ENABLED	Supplies standard deviation noise values to refine the elevation map.
MOTION_INDICATOR	DISABLED	Omitted; redundant during robot motion.
REFLECTANCE_PERCENT	DISABLED	Omitted; ground material profiling is not a requirement for tracking.
NB_SPADS_ENABLED	DISABLED	Omitted; internal diagnostics are excluded from navigation loops.
AMBIENT_PER_SPAD	DISABLED	Omitted; background solar lux tracking provides no obstacle avoidance data.

4.4.3. Multi-Target Detection and Filtering Logic

Because each discrete zone expands as a cone, a single zone can simultaneously hit the edge of a foreground obstruction (Target 1, tracked as index `t1_idx`) and a background boundary (Target 2, tracked as index `t2_idx`). When the sensor queries a flat, uniform surface, all backscattered photons arrive within a single time-of-flight window, causing the sensor to report `nb_target_detected = 1` and proximity data only for Target 1. Under these conditions, Target 2 returns a null marker flag (`'X'`), indicating no second target.

4. Hardware Design and Firmware Implementation

To prevent the robot from executing unsafe movements based on ghost walls or optical glare, data filtering was implemented within the processing loop. The filter evaluates the incoming `TARGET_STATUS` byte for every zone, applying the following criteria [13]:

- **Retained Data Fields (Valid Targets):** Status codes 5 and 6 indicate valid range measurements. Status codes 9 and 10 also represent valid ranges, where 9 signifies a valid range with a large pulse (potentially due to a merged target) and 10 means a valid range with no target detected at the previous range window. Status code 12 identifies a target blurred by another due to sharpener effects, while status code 13 indicates a detected target with inconsistent data; because this state appears frequently when there is a second target, it is included as a valid status.
- **Discarded Data Fields (Noise / Faults):** Status codes ranging from 0 to 4, 7, 8, and 11 indicate that the internal sensor math failed, signal strength was weak on the SPAD array, or consistency checks failed. Status code 255 indicates that absolutely no object reflection was detected. The processing loop catches these fault indices and instantly overwrites their distance values with a null marker flag ('X'), ensuring the navigation system ignores the corrupted data.

4.4.4. SPI Peripheral Optimization and Pin Interfacing

Data transfers are managed by SPI peripherals of the STM32F723VET6 MCU. The peripherals are initialized to operate in full-duplex Master mode, processing 8-bit data transfers synchronized on a low-polarity, single-edge clock phase layout.

During initial validation, configuring the bus with an aggressive clock rate (baud rate prescaler = 2) caused immediate system failures, throwing a "SPI communication not detected" initialization error. This high-frequency clock speed degraded the signal transitions. To fix this issue, the clock tree was reconfigured to utilize a conservative division factor `SPI_BAUDRATEPRESCALER_64`. Slowing the communication bus down to this prescaler stabilized the signal edges, establishing error-free communication loops with all sensors.

The `NCS` pins for all six sensor nodes are configured as standard high-speed GPIO outputs in a push-pull configuration with no internal pull-up or pull-down resistors, initialized to a default high logic state. The connection lines are split into two data retrieval configurations:

- **Interrupt-Driven:** The three sensor modules routed to the `SPI_4` channel are tied to dedicated external interrupt pins configured as `GPIO_EXTI` lines. These pins are initialized with a falling-edge trigger mode and no internal pull-up or pull-down configuration. When a sensor finishes an active ranging cycle, it pulls its local interrupt track low, forcing the MCU to execute a data read.

4. Hardware Design and Firmware Implementation

- **Polling-Driven:** The remaining three sensor modules routed to the SPI_1 channel utilize the same physical interrupt pin hardware configuration, but the core firmware handles them via software polling loops instead, querying the lines and reading data arrays only when the register state transitions to a valid data-ready level.

The performance and current consumption differences between these two data-retrieval methods are evaluated and compared in Chapter 5.

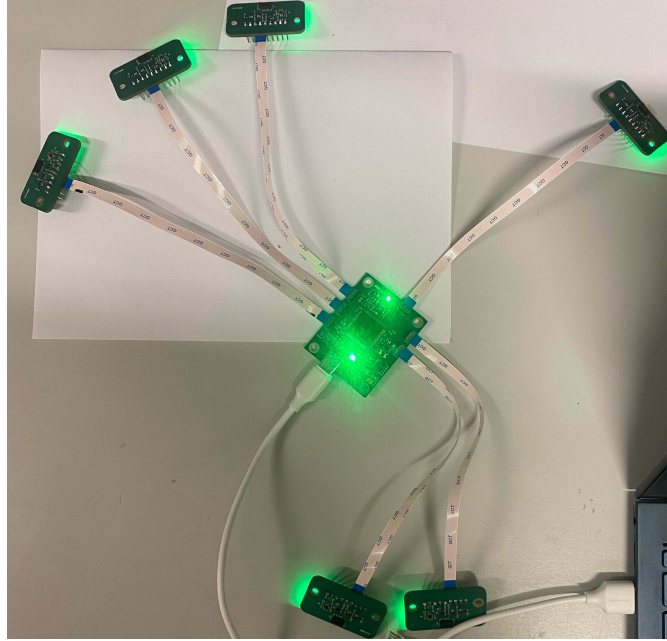


Figure 4.6.: The full module under laboratory testing, displaying the MCU board running the firmware to read data from all six sensor boards

Chapter 5

Results

This chapter presents the empirical validation and performance analysis of the developed perception platform. To evaluate the system under operating conditions, four distinct experiments were conducted: power consumption profiling, testing under different target materials, interference analysis, and indoor clutter mapping. Each section details the experimental setup with the obtained data.

5.1. Power Consumption Measurements

To evaluate the electrical efficiency of the system and validate the calculations developed during the design phase, current characterization was performed. Current waveforms were captured utilizing a Nordic Semiconductor PPK2. The PPK2 executes very accurate continuous sampling to log transient current shifts during various operational states.

5.1.1. Empirical Results

The current scaling profile across different numbers of active sensors is shown in Figure 5.1. The MCU board operation (executing core clock trees, status diodes, and passive networks) draws a baseline current of 101.3 mA. Activating a single dToF node increases the current draw to 149.3 mA. As additional sensors are activated, the power consumption scales linearly, reaching a maximum system load of 400.5 mA when all six sensors actively range simultaneously.

The transient current waveforms shown in Figure 5.1b exhibit periodic peaks and dips. These variations map the operational cycles of the sensor's VCSEL firing. The sharp peaks represent the active transmitting intervals when the laser fires photon pulses. The dips correspond to the subsequent receiving and processing intervals, during which the

5. Results

laser shuts off and the sensor executes mathematical ranging operations. Changes to target profiles or distances do not alter this current profile; the execution cycles of the VCSEL remain uniform (as expected) at 15 Hz regardless of the surrounding environment.

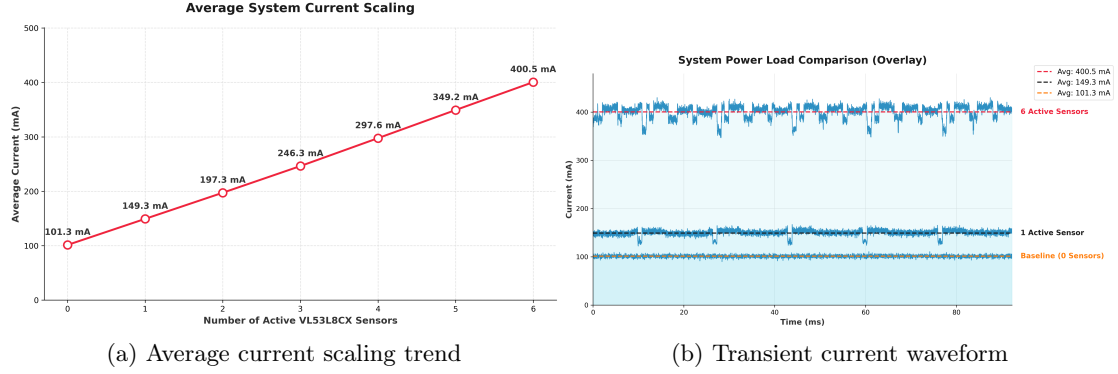


Figure 5.1.: Current consumption profiles: (a) linear consumption scaling mapped across progressive sensor node activations, and (b) transient current waveform comparisons highlighting baseline, single-node, and full six-node module

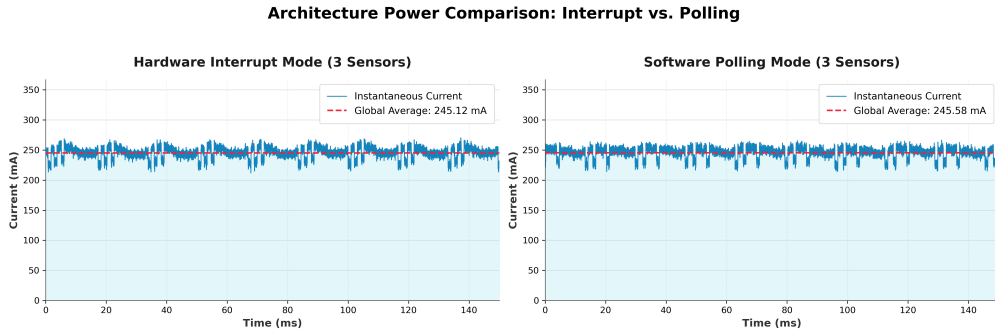


Figure 5.2.: Current comparison tracking three active sensors running under hardware interrupt-driven mode versus software polling loops

Figure 5.2 illustrates that the interrupt-driven architecture draws an average current of 245.12 mA, while the software polling configuration draws a global average of 245.58 mA. The resulting difference (≈ 0.46 mA) is negligible. This near-identical current consumption occurs primarily because the MCU remains continuously in Run mode across both configurations rather than entering a low-power sleep mode. Furthermore, because the continuous power requirements of the active VCSEL dominate the system's total power budget, any minor variations in MCU execution overhead become insignificant.

These measurements align with the estimates calculated in Section 4.2.4. The predicted current of 95 mA for the MCU only plus the current consumption of passives and diodes matches the measured baseline of 101.3 mA, while the predicted full-system load of

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405 mA closely mirrors the measured real load of 400.5 mA. This precise alignment validates the current and power model, confirming that the platform operates safely within its current limits.

5.2. Interference Test

To evaluate the system's susceptibility to mutual interference, two neighboring sensor boards (*Sensor 4_1* and *Sensor 4_3*) were positioned side-by-side, spaced roughly 5 cm apart. Both modules were directed at a uniform, light blue surface located at approx. 45 cm. The objective was to determine whether overlapping 940 nm infrared emissions within the same target area would induce cross-talk, distance degradation, or mutual blindness. The test was executed in two phases: first establishing an isolated baseline with *Sensor 4_3*¹, operating alone, and subsequently activating *Sensor 4_1* to evaluate simultaneous firing of *Sensor 4_3* and *Sensor 4_1*.

5.2.1. Empirical Results

The output comparison during both testing phases is illustrated in Figure 5.3. When operating in isolation, *Sensor 4_3* maps the target surface at a baseline average of approximately 450 mm, with the exception of localized data deviations visible in the lower-left quadrant of the visualization (meaning, upper-right quadrant of the receiver matrix) plot due to a factory protective film left on the lens casing. When both units fire simultaneously into the target zone, the total absence of mutual cross-talk or signal degradation is clearly verified. *Sensor 4_1* (which did not have a protective film) returns a completely noise-free readout of approximately 450 mm for Target 1 across its entire grid, remaining entirely unaffected by the neighboring emitter. Furthermore, while the physical barrier of the protective film causes some primary Target 1 zones on *Sensor 4_3* to drop out to 'X' or false values, during joint operation the true distance metrics successfully map to the Target 2 data slots instead of experiencing optical blindness from the adjacent sensor. This demonstrates that the variations in the baseline chart are purely localized mechanical artifacts of the film.

Due to the uniform target surface utilized for this test, *Sensor 4_1* Target 2 zones capture nothing (as they should). Within the underlying firmware implementation, the target status byte marks these missing returns as a measurement error (in this case - no target detected), which is assigned an 'X' output value. To preserve graphic cleanliness, these null measurements are intentionally omitted from the plots across all experiments unless a valid secondary reflection is captured. This lack of signal distortion confirms that the VCSEL effectively distinguishes its own reflections from neighboring optical emitters,

¹The sensor naming convention follows the format *Sensor X_Y*, where X denotes the active SPI bus (SPI 1 or SPI 4) on the MCU and Y indicates the specific sensor index on that bus. For example, *Sensor 1_2* represents the second sensor on SPI bus 1.

5. Results

validating the layout for multi-sensor integration without the risk of mutual blindness. This operational resilience is further supported by the physical deployment geometry; when distributed around the robot chassis, any overlapping regions of the sensor cones occur exclusively at the outer peripheral edges of the FoV. Because these intersection zones naturally occur at a greater distance than the 450 mm baseline evaluated in this test, the risk of optical cross-talk is reduced even further, keeping mutual exposure far from critical close-range vulnerabilities.

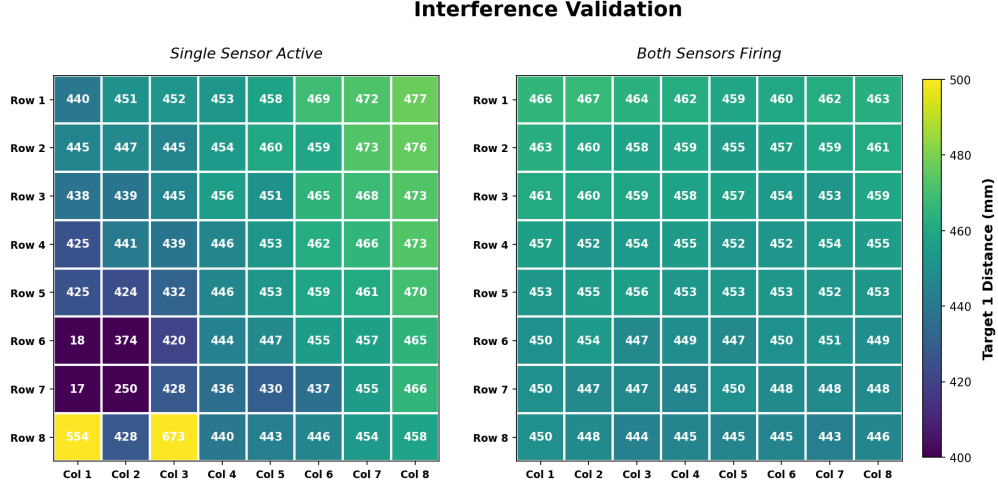


Figure 5.3.: Target 1 distance matrices tracking single-sensor operation versus active dual-sensor firing into a shared target zone.

During the control and calibration phases of this experiment, two anomalies were isolated and analyzed:

- **Lens Blockage:** To isolate a single active emitter while simultaneously simulating environmental dirt, a piece of tape was placed directly on the lens of *Sensor 4_1*. This test revealed a dual tracking behavior across the matrix. In some zones, Target 1 accurately captured the tape at an ultra-close distance of 2 mm, demonstrating the sensor's excellent capability to detect immediate hazards. However, in the majority of the 64 zones, the intense reflections and pixel saturation corrupted the data, causing Target 1 to report false readings ranging from 700 mm to 1300 mm, while Target 2 returned a null marker in most instances and rarely a false reading.
- **Protective Shipping Film Interference:** *Sensor 4_3* exhibited errors, returning negative or very low values for Target 1 on some slots and shifting the true target distance to the Target 2 data slots. This behavior was traced to the factory protective foil on the lens. The film acted as an immediate reflective barrier, redirecting a percentage of the emitter pulse back into the receiving silicon before it could escape. Removing this film eliminated the reflection, returning Target 1 to its true 450 mm coordinate and Target 2 to 'X'.

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These anomalies emphasize the critical requirement for keeping the sensor completely clean on the robot.

5.3. Cluttered Environment Test

To evaluate how the sensor handles realistic scenes, an experiment was conducted in a cluttered indoor environment. The sensor was pointed at a wall with a storage shelf filled with various objects and a chair in front. The objective of this evaluation was to verify the reliability of the dual-target when mapping complex surfaces.

5.3.1. Empirical Results

The resulting 8×8 spatial data matrix is illustrated in Figure 5.4. The middle region cleanly captures the wall, registering a baseline distance of roughly 1630 mm. The higher values approaching 1700 mm at the upper outer boundaries are a result of the sensor’s FoV cone, where the peripheral zones measure a diagonal path to the wall rather than a perpendicular distance.

The lower-left quadrant (Rows 5–8, Columns 1–3) maps the chair positioned in front of the wall. The sensor board was mounted approximately 60 cm above the floor, and it successfully isolated this obstacle, dropping the Target 1 distance readings to 770 mm–862 mm (because the chair was sideways and not facing the sensor directly). Within these same zones, the split light cones allowed Target 2 to concurrently track the wall or other small objects behind the chair at 1479 mm–1590 mm. In the upper-left quadrant, the sensor captures the wall visible above the chair.

The center-right sections of the grid correspond to the multi-tiered shelf. Target 1 values ranging between 1200 mm and 1500 mm map the depths of the various objects arranged on the shelves, while zones showing values around 1630 mm represent open shelf spaces where the background wall is visible. Target 2 appears specifically along the edges of these objects, capturing the exact points where a single measurement zone hits both a foreground item and the back wall.

Notably, two anomalous Target 2 readings of approximately 2100 mm are visible in Row 6 (Columns 5–6). These values exceed the boundary of the wall and are false data, likely caused by low phase returns from an absorptive black object on the shelf. Ultimately, these results demonstrate that despite minor optical anomalies, the dual-target system functions reliably in complex environments, capturing accurate surface profiles.

5. Results

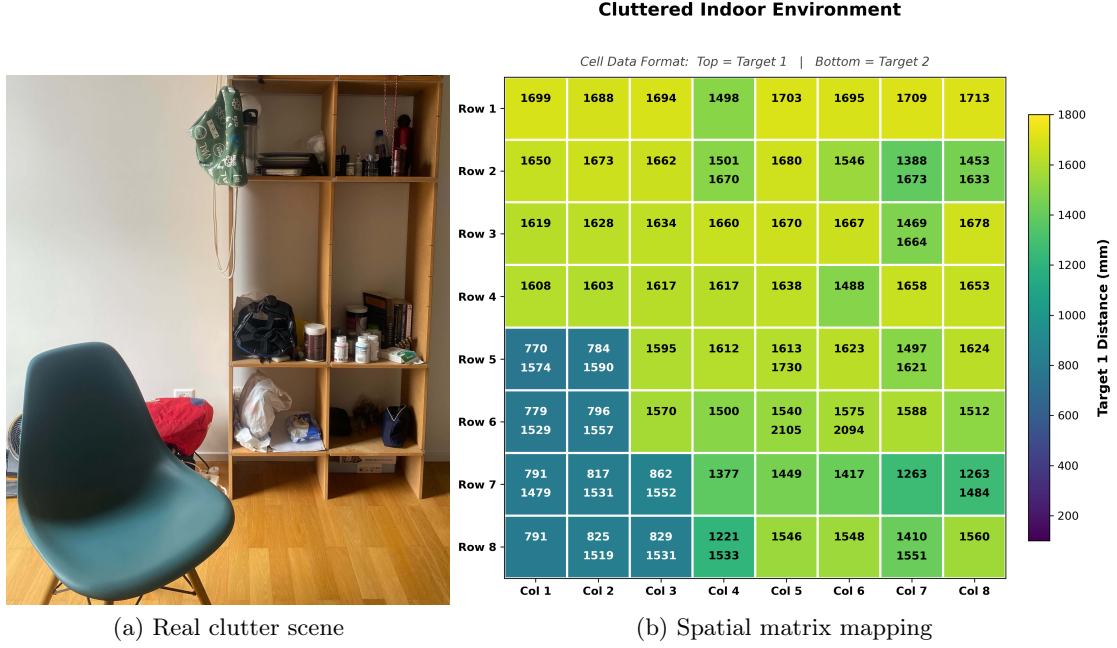


Figure 5.4.: Cluttered scene detection: (a) real clutter scene, and (b) matrix data output highlighting secondary targets along shelf edges

5.4. Material and IR Absorption Test

To characterize how different surface and optical properties impact the dual-target tracking, an absorption and reflectivity experiment was conducted. The experimental setup established a baseline reference target - a flat, matte light blue wardrobe (same as for the interference test) positioned at approximately 770 mm from the sensor face. This wardrobe served as a common background framework for all subsequent evaluations. Three test materials were then introduced into the center FoV in front of this common backdrop: a glass pane (high-transmittance), a white paper sheet (high-reflectance), and a black T-shirt (high-absorption). Finally, qualitative close-up trials were conducted at 20 cm to analyze the sensor's behavior when the same black and white targets are closer.

5.4.1. Empirical Results

The system baseline tracking the wardrobe is shown in Figure 5.5a. Target 1 establishes a stable tracking matrix at the true 775 mm distance, while the Target 2 channel remains completely blank ('X'), showing a perfect operation of Target 2.

5. Results

The high-transmittance glass pane test is plotted alongside the baseline in Figure 5.5b. This test highlights the performance of the dual-target architecture when tracking transparent hazards. Target 1 successfully captures most of the glass boundary in the center of the FoV, while Target 2 peers through the transparent media to track the background wardrobe. Because the glass allows most of the infrared light to pass through, the primary reflection is weak. Despite this, the platform successfully detects most of the transparent barrier while preserving spatial awareness of the environment behind it.

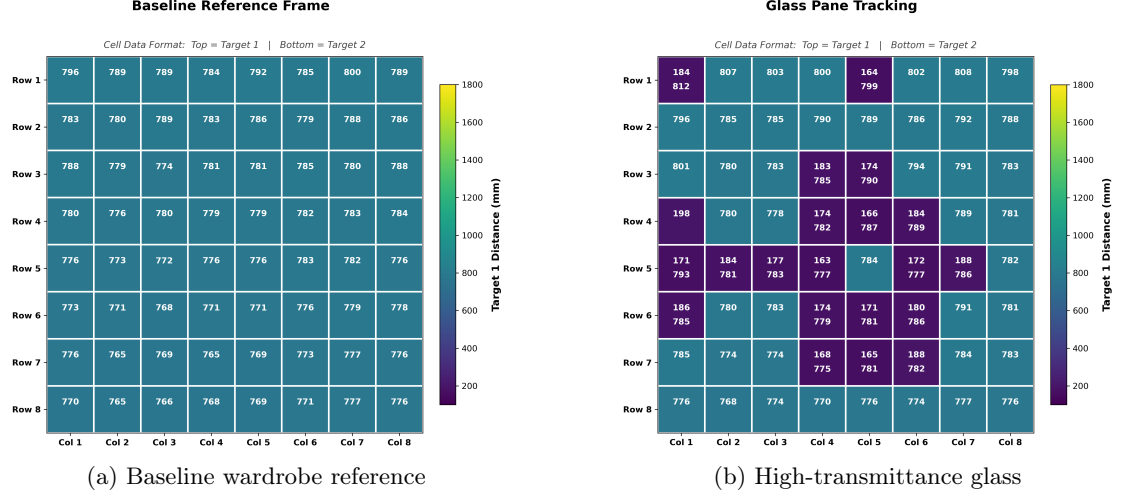


Figure 5.5.: Tracking matrices: (a) baseline measurements of the common wardrobe, and (b) simultaneous detection of foreground glass and the wall behind it

Introducing a high-reflectance white sheet on the baseline demonstrated excellent performance with clean tracking on Target 1, as shown in Figure 5.6a. Because both the background panel and the white sheet reflect many photons, color changes do not alter the primary distance accuracy. The secondary tracking array (Target 2) exhibits near-ideal behavior (i.e., nothing detected because the white sheet is directly on the surface baseline), showing only a few scattered outlier entries across the matrix. These occurrences are a result of micro-bends and minor surface texture variations across the sheet, which induce variations in the infrared beam paths.

Figure 5.6b details the system’s performance when mapping the highly absorptive black target on the common baseline. Target 1 channel successfully isolates the shirt structure across all 64 zones without experiencing blindness. However, because the dark fabric absorbs a significant portion of the emitted 940 nm laser, the returned signal strength drops drastically. Under these low-signal conditions, the Target 2 channel records a cluster of phantom secondary readings ranging between 1000 mm and 1700 mm instead of returning a null value.

5. Results

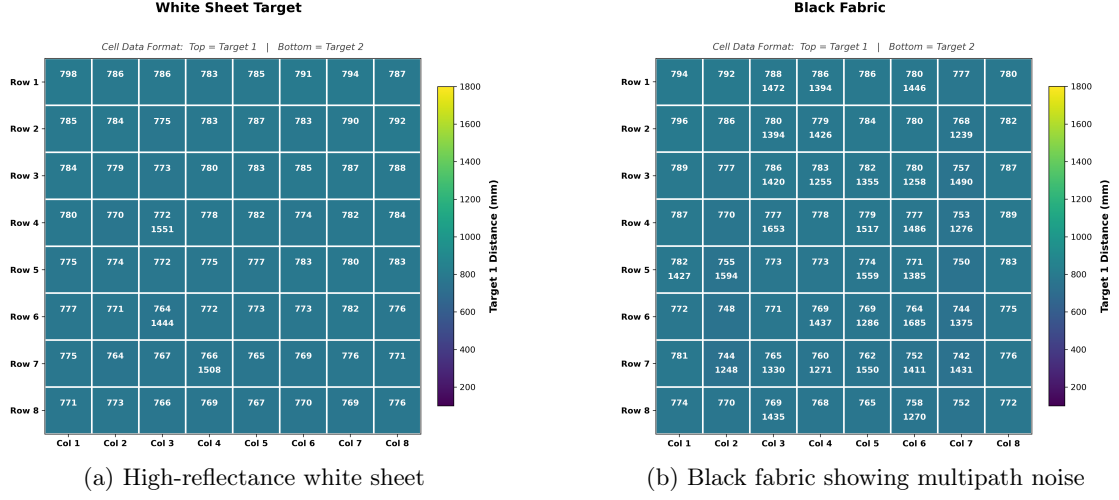


Figure 5.6.: Comparative results for: (a) high-reflectance white sheet with minor surface scattering, and (b) black fabric inducing secondary reflections

During the close-up tests conducted at 20 cm, Target 1 captured both white and black materials much better. For the white sheet evaluation, the minor outliers observed in the longer-distance test are completely gone. The sensor captures the paper on Target 1 while also exposing the wardrobe on the edges and returning a definitive null marker ('X') on Target 2 with zero phantom reflection noise.

The readings for the black fabric are significantly improved compared to the 775 mm configuration. Although the absorptive material still requires a sensitivity boost to maintain a secure lock on Target 1, the shorter distance provides much cleaner signal returns. Consequently, the Target 2 channel is far more stable, displaying a reduced number of false background values.

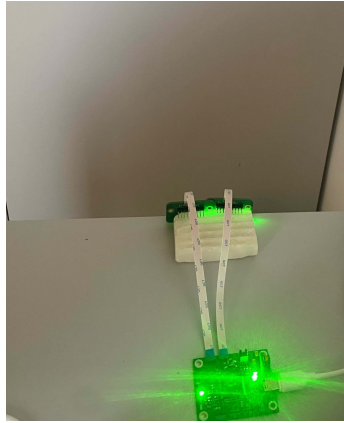


Figure 5.7.: Real interference scene, showing also the uniform target used for baseline

Chapter 6

Discussion and Design Limitations

The empirical evaluations in Chapter 5 demonstrate the stability and performance of the developed perception module. The ToF sensor array achieved good distance tracking accuracy and efficient linear power scaling. Specifically, the electrical profiling verified that the entire module draws a maximum current of 400.5 mA under full operational load, translating to a total power consumption of approximately 2 W. This minimal electrical footprint is advantageous for mobile robotic applications, proving that a network of multiple distributed, non-power-hungry sensors can be deployed without placing a heavy burden on the robot's main battery payload. Furthermore, interference testing confirmed that VCSEL pulse-coding prevents mutual blindness, allowing adjacent nodes to fire simultaneously into overlapping tracking zones without signal degradation.

Overall, the sensor board demonstrates reliable performance in capturing both high-reflectance and high-absorption targets on the primary channel (Target 1). However, depending on the operational distance, the system can display secondary false data on the Target 2 channel, especially when encountering dark objects. When mapping the highly absorptive black fabric target, the returned signal strength drops drastically because the material absorbs a significant portion of the emitted 940 nm laser. My assumption is that to compensate for this signal drop, the sensor's internal algorithms maximize its electronic gain. This hyper-sensitivity is hypothesized to cause the receiver to capture weak reflections bouncing off the adjacent table surface where the sensor board was mounted. This provides a probable explanation for why the Target 2 data is filled with a cluster of phantom readings ranging between 1000 mm and 1700 mm instead of returning a null state.

For a mobile robot deployment, these material-dependent anomalies highlight the necessity of prioritizing Target 1 data within the navigation pipeline. By focusing more on the primary tracking coordinates, the framework can ensure that it effectively ignores these fake background obstacles to maintain safe obstacle avoidance capabilities.

6. Discussion and Design Limitations

Interestingly, the primary tracking channel remained remarkably stable at close range, exhibiting fewer phantom values on Target 2 as the black fabric target was moved closer to the sensor in the performed qualitative tests at 20 cm with the same black and white targets. This behavior directly contrasts with the findings of Caroleo, where the tracking reliability of a black vinyl target dropped sharply at close range (yielding just 18.8% reliability at 25 cm compared to 34.4% at 60 cm). This difference highlights the profound impact of the characteristics of the target material. I believe that the shiny, smooth surface of the black vinyl causes a mirror-like reflection that can easily bounce the laser beam completely away from the receiver at close distances if there is any misalignment. In contrast, the rough texture of the black fabric causes a diffuse reflection, scattering the photons in all directions so that some stream of light always returns back toward the receiving array.

Furthermore, this phenomenon can be analyzed alongside the sensor’s specifications; the VL53L8CX datasheet notes that the device’s internal histogram algorithms are designed to ensure cover glass crosstalk immunity beyond 60 cm. For a highly absorptive, shiny target like vinyl, the weak signal reflected by the target at 25 cm is easily overwhelmed by the internal cover glass noise, causing high measurement rejection rates. Conversely, I assume that because of the diffuse nature of the black fabric, a much higher concentration of photons returns at close range. This return signal is hypothesized to be strong enough to overcome the internal crosstalk threshold, preserving measurement stability even at close distances.

Most importantly, the operation of the whole module throughout all these evaluations proves that the hardware and firmware development of this thesis functioned successfully.

6.1. Hardware Design Limitations

A critical evaluation of the post-fabrication prototype isolated several design vulnerabilities and layout routing errors that must be addressed in future iterations.

6.1.1. Board Sizing and Component Footprint Trade-offs

The design directive required minimizing the MCU board dimensions as much as possible, though no fixed boundaries were targeted. To facilitate manual assembly, passive components were implemented using 0603 and 0805 footprints. These can be optimized down to a compact 0402 standard to reduce footprint area. Additionally, the integrated DC-DC regulator utilized a bulky through-hole module package; replacing this component with a surface-mount device alternative would significantly minimize the vertical profile and overall board layout area. Finally, although the sensor board already utilizes

6. Discussion and Design Limitations

0402 components, optimizing the component placement would allow for more compact board area.

6.1.2. Component Assembly

An assembly mismatch occurred with the reset button. The board was designed for a horizontal-switching configuration, but a vertical-switching component was mistakenly sourced. This pinout conflict was resolved by cutting off two of the opposite legs and placing the button diagonally across the PCB pads, successfully restoring manual reset capabilities.

6.1.3. SPI Clock and INT Routing Error

The most severe hardware error occurred during schematic design, when an interrupt signal line and the `SPI_CLK_1` line were accidentally swapped, but the firmware remained correct. This caused the hardware clock line to be routed to an MCU pin that lacked SPI peripheral support, preventing communication initialization.

To fix this issue without refabricating the board, a physical layout modification was executed. The copper traces for both the misrouted interrupt line and the `SPI_CLK_1` line were cut, and enameled magnet wire was soldered to re-route both the clock and interrupt signals to their correct respective MCU pins.

Furthermore, within the same schematic design, an additional routing conflict occurred on the sensors on the `SPI_1` bus, where an interrupt line (`INT`) was accidentally swapped with a no-chip-select line (`NCS`) on one of the sensor nodes. Due to limitations regarding external interrupt mapping (the MCU's inability to configure concurrent hardware interrupts on different GPIO ports that share the same pin index number), a hardware-driven interrupt routine could not be allocated for this specific node. Consequently, software polling had to be implemented to retrieve data from this sensor. To preserve architectural consistency across the communication bus, the firmware was standardized by switching all three sensors on the `SPI_1` bus to software polling mode.

Conclusion and Future Work

This thesis highlights the development and fabrication of a custom MCU board architecture configured to orchestrate an array of six VL53L8CX ToF sensor nodes, with each node operating at a continuous 15 Hz data acquisition rate. Experimental evaluation characterized the platform’s ranging behavior and tracking data under challenging conditions, focusing on target profiling across transparent glass and highly absorptive materials. Additionally, power characterization successfully validated the analytical models, mapping a precise linear consumption scaling trend from a 101.3 mA bare-board baseline up to a 400.5 mA full-system load.

Although the perception platform proved functionally viable, testing also exposed several layout and routing limitations. Future design iterations must address them to enhance system robustness and structural compactness. Beyond refining the hardware, subsequent efforts will focus on the mechanical integration of the sensor array onto the robot chassis, alongside the ROS integration of the streaming proximity data to enable dynamic obstacle avoidance.

Appendix	A
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Task Description



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich



Task Description for a Bachelor Thesis on

Integration of Distributed dToF Sensors on a Quadrupedal Robot

at the Department of Information Technology and
Electrical Engineering

for

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Professor:	Prof. Dr. Florian Dörfler
Handout Date:	23.03.2026
Due Date:	01.07.2026

Project Goals

As part of the Robodog Flagship Project, this bachelor thesis aims to design and integrate a first distributed direct Time-of-Flight (dToF) sensing module on a small-scale quadrupedal robot. The long-term goal is to enable dense, lightweight, and power-efficient depth sensing for robust autonomous navigation.

Representative benchmark scenarios include the autonomous navigation track of the ICRA Quadruped Robot Challenges (QRC) [1], where quadrupedal robots must traverse uneven terrain reliably and efficiently. Robust exteroceptive sensing is therefore essential.

The robot currently relies on elevation mapping [2], a 2.5D terrain representation where each grid cell stores the estimated ground height. While computationally efficient and well suited for legged locomotion, elevation maps strongly depend on sensor placement and depth quality.

Recent work has demonstrated the importance of exteroceptive sensing for resource-constrained quadrupeds [3]. However, current systems typically rely on a small number of larger depth sensors, limiting placement flexibility and introducing blind spots.

Modern miniature dToF sensors such as the ST VL53L8CX (8×8 zones) and VL53L9CX (54×42 zones) offer a compact and energy-efficient alternative. This project represents a first step toward distributed dToF sensing by developing a scalable sensing module that integrates multiple miniature ToF sensors with embedded processing and communication to the onboard computer.

The objective is to design a compact multi-sensor PCB module, implement embedded firmware and a communication protocol, and integrate the resulting depth data into the existing ROS-based perception pipeline.

Tasks

The project is structured into three main phases.

Phase 1 (Week 1–5)

1. Familiarize yourself with the Robodog hardware architecture, power system, and existing perception pipeline.
2. Review direct Time-of-Flight sensing principles and analyze the ST VL53L8CX sensor in detail.
3. Define system requirements, including power regulation, communication strategy to the onboard computer, and mechanical constraints.
4. Design the first PCB revision (Rev1) integrating one or two VL53L8CX sensors and a suitable microcontroller.
5. Order the Rev1 boards from an external manufacturer at the end of this phase.

Phase 2 (Week 6–10)

1. Evaluate the VL53L8CX and, if available, the VL53L9CX sensors using evaluation boards, including characterization of range, field of view, and noise properties.
2. Develop embedded software for sensor communication and structured data transmission to the onboard computer.
3. Perform bring-up and validation of Rev1 after manufacturing.
4. Integrate the sensing module into the ROS-based perception framework and verify reliable data streaming.
5. Based on the results of Rev1, design a refined second PCB revision (Rev2) addressing layout improvements or identified issues.

Phase 3 (Week 11–14)

1. Validate and finalize the Rev2 hardware (if fabricated), or consolidate the Rev1 design if no revision is required.
2. Mechanically integrate the sensing module at a selected location on the robot.
3. Integrate the depth measurements into the existing elevation mapping framework and perform qualitative evaluation.
4. Conduct experimental validation on the quadrupedal robot.
5. Document the methodology, hardware design, and experimental results, and prepare the final thesis report and presentation.

Milestones

- Completed PCB design and successful board bring-up.
- Reliable communication between sensing module and onboard computer.
- Functional ROS integration of dToF measurements.
- Mechanical integration and experimental validation on the robot.

Project Organization and Grading

During the thesis, students will gain experience in the independent solution of a technical-scientific problem by applying the acquired specialist and social skills.

The grade is based on the following: Student effort; thoroughness and learning curve; achieving qualitative and quantitative results with a scientific approach; supporting

practical findings with theoretical background and literature investigations; final presentation and report; documentation and reproducibility. All theses include an oral presentation, a written report and are graded. The report and presentation need to have publication grade quality to achieve a good grade. Students are graded based on the official ITET grading form¹.

For students of IIS (Prof. Benini) a special grading scheme exists, please contact your supervisor for details there. Before starting, the project must be registered in myStudies and all required documents need to be handed in for archiving by PBL.

Laboratory Rules

The students agree to follow the lab rules set by PBL staff, for detail please contact us. The most important points are:

- All ETH safety regulations need to be followed², in addition to ones given by PBL staff
- No device in the lab is used without introduction by your supervisor or PBL staff
- No device leaves the lab without being officially borrowed, this is done by PBL staff and needs your Legi.
- Any damage to devices or tools needs to be reported immediately to PBL staff.
- The Lab-desk is clean and free for others after you finished your task, or when you take longer breaks. All tools are correctly sorted into their drawers/cupboards when you leave

Weekly Report

There will be a weekly report/meeting held between the student and the assistants. The exact time and location of these meetings will be determined within the first week of the project in order to fit the students and the assistants schedule. These meetings will be used to evaluate the status and document the progress of the project (required to be done by the student). Beside these regular meetings, additional meetings can be organized to address urgent issues as well. The weekly report, along with all other relevant documents (source code, datasheets, papers, etc), should be uploaded to a clouding service, such as Polybox and shared with the assistants.

¹<https://ethz.ch/content/dam/ethz/special-interest/itet/department/Studies/Forms/Grading%20Form.xlsx>

²<https://ethz.ch/staffnet/en/service/safety-security-health-environment/sicherheit-in-laboren-und-werkstaetten/laborsicherheit.html>

Project Plan

Within the first month of the project, you will be asked to prepare a project plan. This plan should identify the tasks to be performed during the project and sets deadlines for those tasks. The prepared plan will be a topic of discussion of the first week's meeting between you and your assistants. Note that the project plan should be updated constantly depending on the project's status.

Final Report and Paper

PDF copies of the final report written in English are to be turned in. Basic references will be provided by the supervisors by mail and at the meetings during the whole project, but the students are expected to add a considerable amount of their own literature research to the project ("state of the art").

Final Presentation

There will be a presentation (15 min presentation and 5 min Q&A for BT/ST and 20 min presentation and 10 min Q&A for MT) at the end of this project in order to present your results to a wider audience. The exact date will be determined towards the end of the work.

References

Will be provided by the supervisors by mail and at the meetings during the whole project.

Place and Date 17.02.26

Zürich

Signature Student



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Appendix **B**

Declaration of Originality

Disclosure of Generative AI and Assistive Technologies

In accordance with the guidelines of ETH Zurich, the following table details the usage of Artificial Intelligence and assistive technologies during the development of this thesis. Chat histories, code assets, and prompt records are preserved in the main repository: https://git.ee.ethz.ch/pbl/FS2026/mihail_barnev_641.

B. Declaration of Originality

Tool	Purpose	Own effort/check	Relevant sections	Evidence
Gemini 3.1 Pro	Development and refinement of the research question through iterative brainstorming; summarizing related work and extracting key technical details.	Evaluated all suggestions, rejected most of them, and formulated the final research questions and summaries independently.	Chapter 2 & 3	Chat history
Gemini 3.1 Pro	Revision of drafted text to apply more formal, scientific expressions; syntax assistance in generating LaTeX table structures.	Reviewed proposed changes individually and accepted only stylistic adjustments; rejected all content-related suggestions.	Entire work	Chat history
Gemini 3.1 Pro	Code generation for experimental data visualization plots (Python / matplotlib).	Tested generated visualization scripts, identified script execution errors, and adjusted layout logic independently.	Chapter 5	Chat history
Gemini 3.1 Pro	Firmware debugging support, isolating code errors, and advising on code readability improvements.	Evaluated suggestions, fixed bugs manually, and verified logic execution on the MCU hardware; rejected content-related logic changes.	Chapter 4	Chat history
Gemini, Adobe Firefly & Photoshop	Generation, refinement, and editing of the system Architecture Overview figure.	Directed layout generation via iterative prompting of what I expect from the image, and manually finalized borders and text corrections using Photoshop.	Chapter 4, Architecture Overview figure	Chat history
DeepL Pro	Translation of primary source documentation (English ↔ Bulgarian).	Checked translations against original documentation texts and corrected technical terminology manually.	Entire work	—
Grammarly	Correction of sentence grammar, writing style flaws, and typos/punctuation.	Evaluated individual proposals and accepted most.	Entire work	—



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. **In consultation with the supervisor**, one of the following two options must be selected:

- ☐ I hereby declare that I authored the work in question independently, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies¹.
- ☒ I hereby declare that I authored the work in question independently. In doing so I only used the authorised aids, which included suggestions from the supervisor regarding language and content and generative artificial intelligence technologies. The use of the latter and the respective source declarations proceeded in consultation with the supervisor.

Title of paper or thesis:

Integration of Distributed dToF Sensors on a Quadrupedal Robot

Authored by:

If the work was compiled in a group, the names of all authors are required.

Last name(s):

BARNEV

First name(s):

MIHAIL

With my signature I confirm the following:

- I have adhered to the rules set out in the [Citation Guidelines](#).
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date

Zurich, 23.06.2026

Signature(s)

If the work was compiled in a group, the names of all authors are required. Through their signatures they vouch jointly for the entire content of the written work.

¹ For further information please consult the ETH Zurich websites, e.g. <https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html> and <https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html> (subject to change).

Appendix C

Repositories

During the development of this thesis, the custom hardware designs, firmware baselines, and test applications were structured and uploaded to the following repository destinations:

- **Platform Hardware (KiCad):** This directory contains the architecture schematic sheets (MCU & sensor boards) and PCB layout of the MCU board and the unadapted sensor board¹.
- **Firmware (STM32F723):** This directory contains the embedded C codebase running on the STM32F723VET6 MCU².
- **ToF Sensorboard Driver Example:** This workspace provides an implementation and standalone layout of the adapted sensor board³.

¹https://git.ee.ethz.ch/pbl/FS2026/mihail_barnev_641/-/tree/main/BachelorThesis-KiCAD?ref_type=heads

²https://git.ee.ethz.ch/pbl/FS2026/mihail_barnev_641/-/tree/main/Board_firmware_BT_STMF723?ref_type=heads

³https://git.ee.ethz.ch/pbl/FS2026/mihail_barnev_641/-/tree/main/ToF_sensorboard_example?ref_type=heads

C. Repositories

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